

What Quits and Layoffs Reveal About the Business Cycle

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Preliminary Draft

This Version: September 22, 2026

Please click here for latest version: [Latest version](#)

First Version: April 2024

Abstract

To challenge and improve business cycle models of labor markets, we use data on separations into non-employment by destination and reason. Standard models where layoffs are random and quits are selective send too many laid-off workers to unemployment— empirically, a third leave the labor force— and they get the cyclical direction of labor force attachment backward. Adding selective layoffs and random quits resolves these issues and clarifies two channels that affect how labor markets evolve during recessions: (i) labor supply increases as marginal workers hoard their jobs and the displaced keep searching, and (ii) selection in layoffs shifts from marginal toward attached workers. Consequently, output per worker falls through composition rather than TFP, and the welfare cost of recessions is less than half that of the standard model. Through these channels, the volatility of the jobless who want to work is further disconnected from measured unemployment, a prediction we confirm in the data.

JEL CODES: E32, J2, J6

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We thank Justin Bloesch, Domenico Ferraro, Jonathan Heathcote, Kyle Herkenhoff, Zhen Huo, Andre Kurmann, Simon Mongey, Lukas Nord, Xincheng Qiu, Andrea Raffo, Victor Rios-Rull, Richard Rogerson, Abigail Wozniak; participants at FRB Minneapolis, LAEF, NBER, SED, Midwest Macro, SAMM, and other seminars and conferences; and our active community of data users for the discussions and helpful feedback. Sara Canilang provided excellent research assistance and James Holt provided professional editorial assistance. The views expressed are those of the individual authors and do not necessarily reflect official positions of the Federal Reserve Bank of Minneapolis, the Federal Reserve System, or the Board of Governors.

1. Introduction

The workhorse macroeconomic model of gross worker flows interprets the labor market through two channels: flows from employment to unemployment are primarily layoffs, and flows from employment to non-participation are primarily quits. In this framework, following the tradition of [Krusell et al. \(2017\)](#), which we refer to as the “standard model,” layoffs are random shocks that hit all workers with equal probability. Quits, conversely, are endogenous choices made by workers whose net surplus of employment has drifted below zero. This structure has influenced both quantitative business-cycle analysis and how policymakers interpret data: unemployment is associated with displaced workers and non-participation with changes in labor supply.

We confront the standard model, using monthly CPS data from 1978 to 2024. Our series classify every separation into non-employment not just by its destination (unemployment vs. non-participation), but also by its underlying reason (quit vs. layoff). This is an additional dimension the standard model has been asked to match, and it fails on both the levels and cyclicity of these more detailed flows.

On levels, the standard model generates too many quits and too few layoffs because it sends laid-off workers to the wrong destination. In the data, 35.5% of laid-off prime-age workers move directly into non-participation. The standard model sends only 22.1%. To manufacture the missing flows out of the labor force, the model must rely heavily on quits, requiring a quit rate 50% larger than empirical targets. Simple recalibration cannot resolve this tension: raising the value of non-market time would successfully push more laid-off workers toward non-participation, but it would simultaneously inflate the already-excessive quit rate.

On cyclicity, the standard model mischaracterizes labor supply dynamics. Empirically, quits fall during recessions, and laid-off workers are more likely to stay in the labor force. The standard model predicts the exact opposite for quits and understates the increased attachment of laid-off workers. This failure is a direct consequence of assuming that all quits are endogenous and all layoffs are random. Because random layoffs strike average-productivity workers, the standard model requires a drop in aggregate productivity to replicate the cyclical fall in output per worker. This productivity drop depresses wages, which in turn strongly incentivizes workers to quit. Furthermore, to match the baseline level of non-participation using only endogenous quits, the model requires large, persistent shocks. It is this persistence that makes workers who are thinking about quitting less sensitive to the temporary decline in the job finding rate (job hoarding) because they do not expect to reenter the labor force soon. Similarly, since all layoffs are random, the decision to stay in the labor force compares

the cost of searching when job finding rates are low to an increase in the unemployment benefit generosity/length, if applicable. This mechanism is quantitatively too weak to match the data, which signals that recessions actually shift layoffs toward higher-surplus, firmly attached workers.

We demonstrate that introducing selective layoffs and random quits allows the model to successfully match reason and destination flows over the business cycle. Because many laid-off workers exit the labor force, layoffs must be partially selective, meaning they target low-surplus workers near the participation margin. Conversely, quits must be partially random, driven by life events (e.g., disability, family circumstances) that place workers far from the participation margin and render them unresponsive to the business cycle. This distinction provides the degrees of freedom necessary to isolate and correctly measure the subset of quits that actually respond elastically to economic downturns.

We add both ingredients to an otherwise standard Aiyagari economy featuring three labor market states. We model selective layoffs via a cyclically varying continuation-cost shock, which endogenously forces out low-surplus workers. Exogenous quits are modeled as a cyclically invariant participation-constraint shock that drives persistent exits. Estimating the model via simulated method of moments, we target the levels and volatility of our new reason and destination series, alongside standard macro aggregates. With these modifications, the model successfully replicates the composition of separations and captures the countercyclical rise in labor force attachment.

Getting the composition of separations right clarifies our understanding of who loses their jobs over the cycle. In normal times, separations are highly selective: 66% of quits and 31% of layoffs occur to low-surplus workers. During recessions, these selective shares collapse to 19% and 18%, respectively. As marginal workers cling to their jobs, separations become more random, reaching up the surplus distribution to strike firmly attached workers. This cyclical de-selection explains why laid-off workers are less likely to exit the labor force during downturns. It also replicates the finding in [Mueller \(2017\)](#) that the separating pool shifts strongly toward high-productivity workers in recessions, a fact the standard model cannot generate with random layoffs. We use this as a test of the model and find our quantitative predictions for the size of this shift in selection are similar to the estimates in [Mueller \(2017\)](#), even though we use fundamentally different quantitative approaches. Consequently, output per worker falls because of compositional shifts rather than TFP declines, which drops the estimated welfare cost of a recession by more than half relative to the standard model (from 1.55% to 0.63% of lifetime consumption).

Matching the composition of separations correctly also changes the role of labor supply in the model. During recessions, marginal workers hoard their jobs (quits fall), displaced workers

stay attached, and non-participants begin searching. This countercyclical labor supply actually stabilizes employment while amplifying measured unemployment. This countercyclical response causes employment to fall *less* (by 1.0 percentage point at the trough) and measured unemployment to rise *more* (by 0.3 percentage points). The standard model misses this pattern because it features the wrong sign of the labor supply response.

Our remedied model also reveals that the volatility of unemployment over the cycle is much larger than the volatility of unmet labor supply, or those who want to work but don't have a job. The majority of hires from non-employment originate in non-participation, and the share of marginal workers in this pool becomes much smaller during recessions, owing to the forces driving countercyclical labor supply in the model. We use the model to measure the share of the population that is non-employed but wants to work (WTW) as those with values of employment exceeding the value of their current non-employed state, agnostic of whether they are unemployed or non-participants. This stock is six times the unemployment stock in normal times and a third as cyclical in our baseline calibration to the Global Financial Crisis (GFC) era. The wedge between the two is policy and labor supply: in the GFC, the UI extension alone moves the unemployment rate by 0.3 percentage points while moving WTW by a tenth of a point. The unemployment rate is a measure of who is *searching*, and search responds to benefits; WTW is a measure of who would *work*, and that is the object that matters for slack. This prediction offers a great opportunity for validation of our model, and we show the cyclicity is in line with a direct empirical counterpart we construct using questions on wanting to work in the Current Population Survey. It also reinforces the advantages of using a structural model to track slack.

We conclude by estimating the model across two additional episodes. The 1981–82 recession is an era of pure displacement, where job loss drove the unemployment spike, UI was less generous, and selective layoffs fell sharply. In contrast, the tight labor market of 2022–23 represents a macroeconomic boom masked by a substantial decline in labor supply, demonstrating that the small drop in the official unemployment rate severely understated how tight the labor market had truly become.

Literature Review Our paper contributes to the empirical and theoretical literature that examines labor market flows between employment, unemployment, and non-participation.

Labor Market Flows and Transitions. A rich literature following work by [Abowd and Zellner \(1985\)](#) and [Blanchard and Diamond \(1990\)](#) has analyzed gross flows and transition rates between the labor market states. This body of research aims to understand the evolution of labor market flows across time, cross-sections, and business cycles. [Shimer \(2012\)](#)

utilizes flow data on employment-unemployment transitions to construct job-finding and job-loss probabilities, assessing their relative importance to unemployment rate fluctuations. Similarly, [Elsby et al. \(2015\)](#) and [Elsby et al. \(2019\)](#) employ data on flows between employment, unemployment, and non-participation to analyze the contribution of each flow to labor force participation rate fluctuations. Others, such as [Garibaldi and Wasmer \(2005\)](#), [Krusell et al. \(2017\)](#), [Cairó et al. \(2022\)](#), [Qiu \(2022\)](#), and many more have used gross flows data to inform macroeconomic models of labor markets. Our contribution to this literature is to discipline these models with the composition of separations by reason and destination, and to show that the discipline is binding: the workhorse model cannot match it without selective layoffs and random quits.

Closest on the data side is [Simmons \(2023\)](#), who shows in SIPP and the UK Labor Force Survey (LFS) that allowing for heterogeneity in the reason for separation raises the measured contribution of separations to the ins of unemployment via a Shimer-style flow accounting. We reach a complementary conclusion using a structural model, and add the destination margin.

Marginal Workers and Labor Force Attachment. Our focus on marginal workers connects to literature examining labor force attachment. [Jones and Riddell \(1999\)](#) and [Jones and Riddell \(2006\)](#) identify a group of non-employed individuals who report wanting work but are not actively searching. This “waiting” group has transition rates to employment that are higher than other non-participants’ but lower than those of the unemployed.¹ More recently, [Morchio \(2020\)](#), [Gregory et al. \(2025\)](#), and [Ahn et al. \(2023\)](#) identify marginal groups as individuals who flow often between states. We complement this work through a different identification approach, using quits and exits after layoff, to track this distribution closer to real time, instead of relying on work histories.

Our analysis of marginal workers also relates to [Barnichon and Figura \(2015\)](#), who demonstrate that compositional changes in the unemployed population—including shifts in labor force attachment—substantially affect aggregate matching efficiency. Similarly, [Hornstein et al. \(2014\)](#) develop a framework where non-participants’ heterogeneous labor force attachment influences aggregate job-finding rates.

Selection in Quits and Layoffs. [Gibbons and Katz \(1991\)](#) offer empirical evidence for selection in layoffs, showing that workers displaced by plant closings face shorter unemploy-

¹ [Flinn and Heckman \(1983\)](#) test whether unemployment and non-participation are behaviorally distinct labor force states and find that unemployed workers have significantly higher job-finding rates than non-participants. Building on this, [Krueger et al. \(2014\)](#) examine the long-term unemployed and find that many become marginally attached to the labor force, cycling between active search and non-participation.

ment durations and smaller wage losses than those laid off for other reasons. More recently, [Mueller \(2017\)](#) finds that laid-off workers have substantially lower pre-displacement wages than those who quit, which supports the selection hypothesis. [Chodorow-Reich and Karabarbounis \(2016\)](#) document cyclical patterns in the opportunity cost of employment, and [Mui and Schoefer \(2021\)](#) show that selection into unemployment varies systematically over the business cycle. Our model nests both selective and random separations on each margin and lets the data determine the mix, in each phase of the cycle.

Unemployment Dynamics and Slack. Our work contributes to understanding unemployment movements not directly related to labor market slack. [Ahn and Hamilton \(2020\)](#) decompose unemployment fluctuations into flows related to entry, exit, and duration, finding significant variation unrelated to aggregate demand.

We contribute to the “unemployment volatility puzzle” literature pioneered by [Shimer \(2005\)](#) and further explored by [Hagedorn and Manovskii \(2008\)](#), [Ljungqvist and Sargent \(2017\)](#), and [Mitman and Rabinovich \(2019\)](#). We do so by showing that layoffs are more frequent and less cyclically volatile than flows from employment to unemployment, and that labor supply appears to be countercyclical on the margin.

[Mukoyama et al. \(2018\)](#) document countercyclical search intensity, suggesting that labor supply behavior—rather than just demand factors—shapes unemployment fluctuations. The role of unemployment insurance in distorting unemployment measures of slack has been investigated by [Rothstein \(2011\)](#) and [Farber and Valletta \(2015\)](#), who find that UI extensions modestly increase unemployment duration but primarily through reduced labor force exits rather than reduced job finding. This aligns with our finding that marginal workers are more likely to report as unemployed when benefits are more generous.

Cyclical Labor Supply. Most directly related to our central finding is the contemporaneous work of [Graves et al. \(2023\)](#), who estimate the response of labor-market flows to high-frequency monetary-policy shocks. They find, as we do, that labor supply cushions the employment decline through the response of quits to a falling job-finding rate, in a model in the tradition of [Krusell et al. \(2017\)](#).² We, however, obtain estimates about the role of labor supply that are less than half as large: the employment decline would be about a third larger

²Our CPS quit and layoff series were developed independently. We apply standard and publicly documented procedures for the panel component of the CPS: the survey’s long-standing reason-for-separation questions for the unemployed (harmonized in IPUMS; see [Flood et al. \(2023\)](#)) and for non-participants (collected since 1967 and long analyzed in the literature, e.g. [Flaim \(1969\)](#), [Flaim \(1973\)](#), [Schwab \(1974\)](#), [Bednarzik and Klein \(1977\)](#), and [Gellner \(1975\)](#)), with longitudinal matching following [Madrian and Lefgren \(1999\)](#). The series we construct were the first of their kind made publicly available; they classify every separation by reason and destination and are updated monthly.

without the labor-supply response in our model, against roughly 80% larger in theirs. The difference turns on how each model interprets the central fact that nearly 40% of laid-off workers exit the labor force. Because [Graves et al. \(2023\)](#) allow only random layoffs, their model must read this fact as evidence that 40% of *all* workers are marginal: they sit close enough to the participation margin to exit upon displacement, and it also attributes the entire cyclical fall in exits after layoff to a pure labor supply response. That delivers a very high labor-supply elasticity. Our model instead attributes laid-off workers’ exit to both labor supply and selection: layoffs fall disproportionately on low-surplus workers, who are the ones likely to exit, and part of the cyclical movement is a shift in *who* is laid off rather than a change in workers’ participation choices. Critically, our mechanism is validated by estimates of cyclical selection by [Mueller \(2017\)](#), which our model succeeds in replicating, despite not targeting, in the estimation.

2. Data Methodology and Findings

2.1. The Classification

Most quantitative work on gross worker flows uses the nine transition rates between employment (E), unemployment (U), and non-participation (N). The usual convention has been to treat flows from employment to unemployment (EU) as layoffs and flows from employment to non-participation (EN) as quits. We show that this classification is not accurate. Both destinations contain both kinds of separation, and the mix moves systematically over the business cycle. We therefore work with monthly series that classify every separation into non-employment by its reason as well as its destination, splitting the two separation flows into four: employment to unemployment due to a quit (EUQ) or a layoff (EUL), and employment to non-participation due to a quit (ENQ) or a layoff (ENL). The destination flows are the sums

$$f_{EU} = f_{EUQ} + f_{EUL} \quad \text{and} \quad f_{EN} = f_{ENQ} + f_{ENL}, \quad (1)$$

and total separations into non-employment by reason are

$$\text{Quits} = f_{EUQ} + f_{ENQ} \quad \text{and} \quad \text{Layoffs} = f_{EUL} + f_{ENL}. \quad (2)$$

Table 1 shows the resulting matrix.

The series are seasonally adjusted, span 1978 to 2024, and are constructed and validated in [Ellieroth and Michaud \(2026\)](#), which documents the underlying CPS questions, the longitudinal matching of respondents across consecutive months, robustness checks, and validation against the PSID and the SIPP. Appendix A reports the subset of those checks that are important for the data moments in this paper. We restrict our sample to the prime-age

From		To	
	E	U	N
E	f_{EE}	$f_{EUQ} + f_{EUL}$	$f_{ENQ} + f_{ENL}$
U	f_{UE}	f_{UU}	f_{UN}
N	f_{NE}	f_{NU}	f_{NN}

Table 1: Our contribution to the standard approach

population, those 25–55. The design of the questions we use is such that they were never intended to capture retirees.

2.2. What Counts as a Quit and What Counts as a Layoff

The classifications fo quits and layoffs are derived from questions that ask the unemployed and non-participants about the reason their last job ended. For the unemployed, the CPS asks why the respondent became unemployed, and the answer is harmonized in IPUMS for the full sample period. We classify job losers, those on layoff, and those whose temporary job ended as layoffs, and job leavers as quits.

For non-participants, the survey asks why the respondent left their last job. We harmonize the answers across the studied time frame, 1978 – 2024, and classify a completed temporary, seasonal or intermittent job and slack work or business conditions as layoffs; and personal or family reasons– including pregnancy, return to school, health, and unsatisfactory work arrangements– as quits.

2.3. Quits, Layoffs, and Where Separated Workers Go

Figure 1 plots the full time series of the monthly quit and layoff rates to non-employment from 1978–2024 as a percentage of total employment for the prime-age population in the United States. On average, about 40% of all separations into non-employment are quits, and the remaining 60% are layoffs. This implies that in any given month, about 0.85% of all employed workers decide to leave their job– that is, to quit from employment to non-employment. The majority of these quits are into non-participation, which make up about 85% on average.

The data we use allow us to make the important distinction as to why an individual flows into unemployment or non-participation. Whereas the previous literature assumed that an EU transition is due to a layoff and an EN transition is due to a quit, we show where layoffs actually go. We can see in Figure 2 that a substantial share of layoffs, 35% end in non-participation. Clearly, not all layoffs end in unemployment, and similarly, not all movements to non-participation are due to a quit decision, as was previously assumed. This implies that about 35% of workers choose to leave the labor force after being laid off.

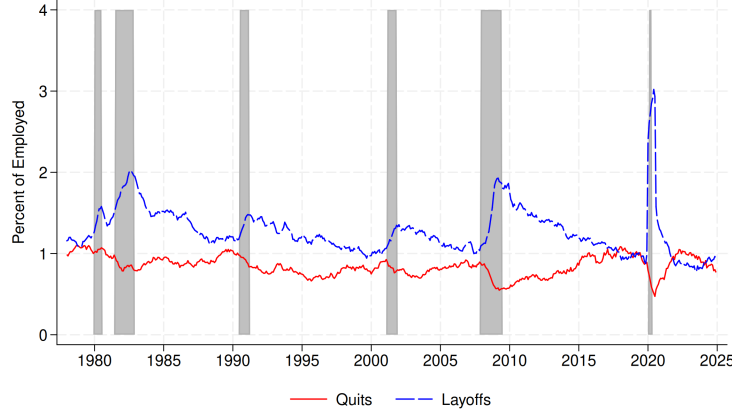


Figure 1: **Monthly quits and layoff rates (as a percentage of employment)** (Monthly seasonally-adjusted data and 6-month centered moving average)

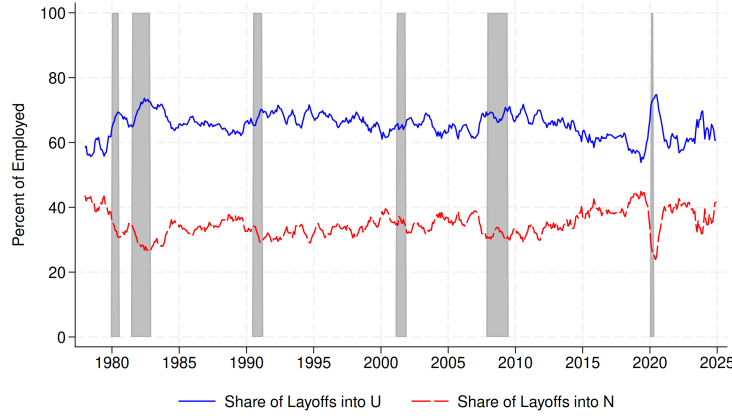


Figure 2: **Share of layoffs by destination (Unemployment or non-participation)**

In total, about 55% of all separations in a given month end in non-participation and are a result of labor supply decisions. Both types of labor supply decisions, quits and leaving the labor force after a layoff, inform about workers' value of employment compared to non-participation.

The 40% of workers who quit in a given month are likely to have a negative employment surplus, or equivalently, the value of non-participation likely exceeds their value of employment. One interpretation of the large number of quits is that many individuals are on the margin of participation; that is, the difference between their value of employment and non-participation is small. Additionally, we observe large transition rates between non-participation and the labor force: about 6% of all individuals in non-participation join the labor force in the average month. These facts suggest that these workers are near the margin

of participation. However, it is unlikely that all quits in the US are due to marginal workers. Some individuals who quit are likely to not join the labor force anytime soon, and remain out of it for a long time. The model will allow us to distinguish between the two types of quits. Understanding the distinction is important for studying the extensive labor supply decisions of these marginal workers, and the factors that drive them.

Moreover, 35% of all layoffs end in non-participation, which implies that these workers likely had a small but positive employment surplus, since they were working. However, the choice of non-participation after a layoff implies that these workers were very close to the margin. The data, which allow for observing the reason and destination of a separation, are crucial for identifying these marginal workers in a model. Laid-off workers' decisions to either stay in unemployment or leave the labor force are central to estimating labor supply responses in the model.

We can see in figures 1 and 2 that both quits and the share of layoffs into non-participation display clear cyclical patterns. The visual inspection shows that both are procyclical; that is, they decrease in recessionary times. Table 2 confirms the visual finding. In times when the unemployment rate is high, the shares of layoffs and quits into non-participation decline. In particular, the share of layoffs into non-participation is strongly negatively correlated with the unemployment rate. One interpretation of this finding is that laid-off workers become

Statistic	Share of Layoffs into N	Share of Quits into N
$\text{Corr}(x, y)$	-0.6367	-0.1528
$\text{SD}(x)/\text{SD}(y)$	0.0231	0.01543

Table 2: **Business-cycle correlation of the share of layoffs and quits into N with the unemployment rate**

more attached to the labor force in recessions as more workers choose to remain unemployed after losing their job— for example, because of an increase in the value of unemployment benefits. Alternatively, this suggests that there is potential selection in the pool of laid-off workers in recessions. In normal times, when the share of laid-off workers exiting the labor force is larger, layoffs might be more targeted toward these small-employment-surplus marginal workers. In recessions, however, layoffs are less targeted, and therefore, the share of layoffs into non-participation declines. The model will help us to identify how much of the decline in the share of layoffs into non-participation is due to a change in the employment surplus versus a change in the composition of laid-off workers.

2.4. The Role of Composition Effects

In this section, we investigate how much of the decline in the share of layoffs into non-participation is due to a change in the composition of laid-off individuals (based on observables) in recessions or changes within different groups of laid-off workers. We do this by computing the counterfactual share of layoffs that flow into non-participation during each of the previous five recessions. Thus, we perform a shift-share analysis similar to [Elsby et al. \(2015\)](#) where share_{it}^{eln} is the weighted average of the fraction of laid-off workers exiting the labor force for different groups of laid-off workers:

$$\text{share}_t^{eln} = \sum_i \omega_{it} \text{share}_{it}^{eln}, \quad (3)$$

where ω_{it} is the share of group i among all laid-off workers in time t . In order to decompose the change in share_t^{eln} , we fix ω_{it} for each group i to the 12-month pre-recession average of ω_{it} , which we call ω_i^C . The counterfactual share of laid-off workers leaving the labor force, share_t^C , is computed as follows:

$$\text{share}_t^C = \sum_i \omega_i^C \text{share}_{it}^{eln}, \quad (4)$$

We calculate three different counterfactual analyses, by gender, race, and education, such that $i = \{\text{men, women}\}$, $i = \{\text{white, black}\}$, or $i = \{\text{high school at most, college, and more}\}$.³

Table 3 compares the actual change in the share of individuals who leave the labor force after job loss to the counterfactual change for each recession. Table 4 shows the percentage change. Both tables show that the composition of the laid-off pool across observable groups

Recession	Actual change in pp	Counterfactual change		
		Gender	Race	Education
Jan 1980 - Jul 1980	-0.0867	-0.0640	-0.0704	-0.0914
Jul 1981 - Nov 1982	-0.0870	-0.0657	-0.0926	-0.0921
Jul 1990 - Mar 1991	-0.0628	-0.0467	-0.0589	-0.0598
Mar 2001 - Nov 2001	-0.0508	-0.0502	-0.0556	-0.0507
Dec 2007 - Jun 2009	-0.1307	-0.1034	-0.1315	-0.1368

Table 3: Actual and counterfactual change (in pp) in the share of layoffs into non-participation

contributes little to the recession decline in the share who exit: the composition based on observables accounts for at most 25% and declined over time. Thus, selection in observables

³Because of the sample size, we do not evaluate interactions of these groups—e.g., black high school men, etc.

Recession	Actual change in percentage	Counterfactual change		
		Gender	Education	Race
Jan 1980 - Jul 1980	-0.0867	-26.18%	-18.80%	+5.42%
Jul 1981 - Nov 1982	-0.0870	-24.48%	+6.44%	+5.86%
Jul 1990 - Mar 1991	-0.0628	-25.64%	-6.28%	-4.78%
Mar 2001 - Nov 2001	-0.0508	-1.18%	+9.45%	-0.19%
Dec 2007 - Jun 2009	-0.1307	-20.89%	+0.6%	+4.67%

Table 4: Actual and counterfactual change (in percentage) in the share of layoffs into non-participation

explains only a part of the higher attachment of laid-off workers to the labor force in recessions. The model will help us understand what accounts for the large share that is not accounted for by observables.

3. Model

We enrich a standard heterogeneous agent model of aggregate labor market dynamics to include features related to labor supply and selection. The foundation is an Aiyagari model, expanded to feature gross worker flows across three labor market states: employment (E), unemployment (U), and non-participation (N), following the tradition of [Krusell et al. \(2017\)](#). To this we add a notion of quits and layoffs, each of which can happen at random or through endogenous choices of individuals, thereby determining the extent of selection in job separation.

3.1. Environment

Time is discrete and the economy is populated by a unit measure of heterogeneous workers. The aggregate state is summarized by a vector $\mathcal{S}$, which includes a business-cycle indicator (Z) and information necessary to determine equilibrium prices (w, r) and provide rational expectations over the future aggregate state.⁴ Labor market frictions, including shocks related to job loss and job finding, as well as some government programs, will depend on the business-cycle indicator.

Preferences. Individuals have preferences, represented by

$$E_t \sum_{t=0}^{\infty} \beta^t [\log(c_t) - \theta(s_t)], \quad (5)$$

⁴TFP and the capital-effective labor ratio determine prices. Our computational approach follows [Boppart et al. \(2018\)](#) MIT-style shocks for expectations, as we detail later in the paper.

where $c_t \geq 0$ is consumption in period t and $s_t \in \{0, 1\}$ is an indicator for whether the individual engages in active job search. The parameter $\theta > 0$ represents the disutility of active search, and $0 < \beta < 1$ is the discount factor. There is no on-the-job-search; individuals choose whether to search or not only when non-employed.

Budget Constraint and Earnings. Individuals cannot borrow, but can accumulate assets a that offer a return $r(\mathcal{S})$. The budget constraint is:

$$\begin{aligned} c_t + a_{t+1} &= (1 + r(\mathcal{S}))a_t + y_t, \\ a_t &\geq 0. \end{aligned}$$

Flow earnings, denoted by y_t , depend on the individual's labor market state. An employed worker's labor earnings are the product of her idiosyncratic labor productivity z_t and the market wage per efficiency unit of labor services $w(\mathcal{S})$, net of labor income tax $\tau(\mathcal{S})$:

$$y_t = (1 - \tau(\mathcal{S}))w(\mathcal{S}) \cdot z_t, \quad (6)$$

As is typical, an individual's idiosyncratic productivity z_t is stochastic and follows an AR(1) process in logs:

$$\log z_{t+1} = \rho_z \log z_t + \varepsilon_{t+1}, \quad (7)$$

where the innovation ε_{t+1} is normally distributed with mean zero and standard deviation σ_ε .

Non-participants engage in home production h , which is constant across individuals and time, and so their flow income is: $y_t = h$.

The model includes an unemployment insurance (UI) program that captures key features of the US system. To be eligible ($e = 1$) for UI, a worker must have previously been employed and experienced a layoff, as defined in the next section. Individuals who voluntarily quit their jobs are not eligible ($e = 0$). To receive benefits, an individual must engage in active search ($s = 1$) and incur the search cost θ . UI benefits have a finite duration. An eligible individual loses their eligibility each period with probability $\mu(Z)$ which allows for UI extensions in recessions.

The value of the unemployment benefit is given by the following formula:

$$b(z) = \begin{cases} b_0 z_t & \text{if } b_0 z_t \leq \bar{b} \\ \bar{b} & \text{otherwise.} \end{cases}$$

where $\bar{b}$ is the maximum benefit amount and b_0 is the replacement rate.

Additionally, there is a means-tested consumption floor c_{min} representing safety net programs. This payment is available to all agents whose period resources $a_t + y_t$ fall below the consumption floor.

Production. The production function is neoclassical. Output is produced by capital services K and effective labor L according to the function $A(Z)K^{\alpha_y}L^{1-\alpha_y}$. Total factor productivity, $A(Z)$, varies according to the aggregate state. The capital stock in production is equal to the total savings across households (i) $K = \int_i a_i$, and labor services are equal to the total efficiency units of labor supplied by the employed. Let $\mathcal{I}_e(i) = 1$ if agent i is employed: then, this is $L = \int_i \mathcal{I}_e(i)z_i$. Capital and efficiency units of labor are rented at their respective marginal products: $r = A(Z)(\frac{L}{K})^{1-\alpha_y}$ and $w = A(Z)(\frac{K}{L})^{\alpha_y}$.

Government Budget. We assume that the government budget clears period by period. In any given period, total revenues equal the labor tax revenue from the employed $TR = \int_i \tau \mathcal{I}_e(i)z_i$, and total expenditures equal total spending on the consumption floor and unemployment benefits $TE = \int_i \mathcal{I}_{ub}(i)b_i + \int_i \mathcal{I}_c(i)c_{min}$ where $\mathcal{I}_{ub}(i) = 1$ for an unemployed individual eligible for unemployment insurance and $\mathcal{I}_c(i)c_{min} = 1$ for an individual receiving the consumption floor.

3.2. Labor Market Transitions and Employment Dynamics

Job Loss: Quits and Layoffs. Employed workers can lose their job each period in four different ways. Some job losses we will classify as quits and others as layoffs. A portion of each of the quit and layoff hazards is totally exogenous and common to all workers. Another portion of each of the quit and layoff hazards is endogenous and selective: in equilibrium, the hazard rates will be higher for workers with a lower surplus from employment. There is no on-the-job search, and so any job loss results in a transition to non-employment.

First, the worker may quit. A selective quit is an endogenous quit that comes about when a worker assesses the value of non-participation or unemployment to be higher than continuing at work. Once separated, he or she can choose whether to search for a new job in unemployment, or to move to non-participation for however long they like. There is also an exogenous shock p that forces a worker into non-participation until the shock goes away. We will call this type of quit “non-selective” or “random” and categorize the worker as participation-constrained, meaning that they cannot make labor market choices and must stay in non-participation. This shock is meant to capture individuals very far from the margin of the labor market for reasons we do not model, such as disability or the ample resources of a second earner. This shock follows a Markov process where $\pi^p(p', p)$ is the probability of moving to state p' from p . No worker exiting employment via a quit is eligible for UI.

Alternatively, the worker may be laid off. As is standard in many labor market models, there is a probability $\delta(Z)$ that a worker is laid off exogenously, and this probability is common across all workers but varies with the aggregate cyclical state (Z). We call these types of layoffs “non-selective” or “random.” The other type of layoff is called “selective” and

is generated by an endogenous response to a shock x . This shock has a random arrival rate of $\lambda_x(Z)$. Upon arrival, the worker must pay a utility cost x in order to remain employed. If they choose not to pay the cost, they are laid off. This is a simple way to capture selective firings of low-surplus workers within the firm. Both a selective layoff and a selective quit target low-surplus workers, and the layoff is completely transitory to the worker. Similar to a match quality shock, it is meant to represent pressures idiosyncratic to the firm that cause it to cut the least productive workers, even though those workers could have a positive surplus elsewhere. An endogenous quit, by contrast, is a separation initiated by a worker who does not have a positive employment surplus at *any* job. While all separated workers have the choice of whether to pay the search cost and enter unemployment or exit to non-participation, it is easy to show that since endogenous quits are due to a persistent decline in a worker's employment surplus, all selective quits will result in a transition to non-participation. By contrast, since endogenous selective layoffs are driven by transitory (one-period) changes in the employment surplus, they can result in the worker choosing either unemployment or non-participation.

The rich set of transitions between labor market states is key to our analysis, and so we summarize them here, providing the clear ranking of average wage and duration of non-employment that will result from each type of flow in equilibrium.

1. **Random Quits (*average wage, longest duration of non-employment*)**: With probability $\pi^p(1, 0)$, a worker is forced to quit into non-participation, where they remain until the shock is removed. These are the quits of non-marginal workers who are far from participating despite changes in policy or their circumstances.⁵
2. **Selective Quits (*lowest wage, medium-long duration of non-employment*)**: Employed individuals choose to quit if the surplus of employment relative to non-employment is negative. These are the quits of marginal workers who are relatively close to participating if policy or their circumstances change.
3. **Selective Layoffs (*medium-low wage, medium-short duration of non-employment*)**: With probability $\lambda_x(Z)$, a worker must pay a one-time utility cost x to continue an employment match. If they refuse, they are laid off. This will endogenously generate layoffs of marginal workers with low employment surplus who are more likely to choose to exit the labor force after the layoff.

⁵In the quantitative model, we provide these agents with flow consumption and fix their assets at point of entry so that they do not factor into our analysis of business-cycles. It is equivalent to removing these flows from both model and data instead.

4. Random Layoffs (*average wage, shortest duration of non-employment*):

With probability $\delta(Z)$, a worker is randomly laid off. This will generate layoffs equally among workers who are strongly attached with high employment surplus, and among those marginally attached with low employment surplus.

We validate the above rankings directly using data from the Panel Study of Income Dynamics (PSID).⁶ Figure 3 plots the distribution of residual earnings in the month before separation for each group. The ordering is the same as our model predicts. Workers who quit to non-participation experience the lowest wages, those laid off into unemployment have the highest, and those laid off into non-participation lie between them.

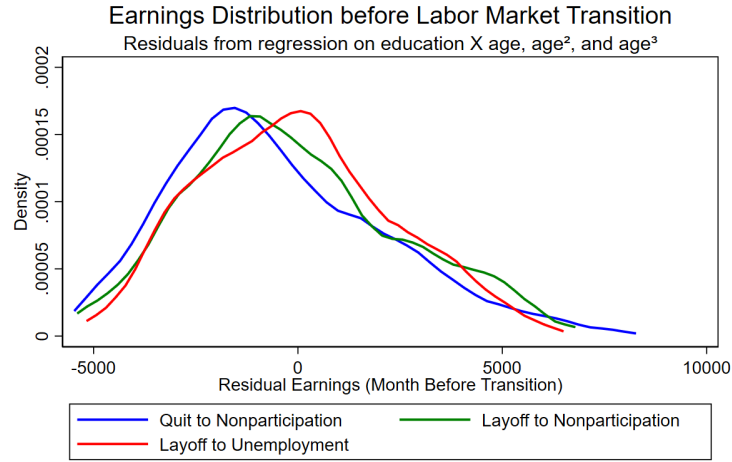


Figure 3: **Distribution of residual earnings in the month before a labor market transition, by type of transition (PSID).** Residuals are from a regression of earnings on education interacted with a cubic in age (see Appendix B.1).

Figure 4 analogously shows the duration of non-employment for the different types. Individuals who got laid off and choose unemployment have the shortest non-employment duration, followed by layoff into non-participation, and quits into non-participation. Again, this is the same ordering as we get from the model.

Finally, a measure R_d of agents die each period and are replaced by an equal measure of newborn agents. The newborn agents begin non-employed and are endowed with assets equal to the average holdings in the economy. Deaths are not counted in labor market flows in the model (and they are not counted in the flows in the data). The newborn agents are counted as originating from non-participation. If they choose unemployment in their first period, then that will be counted as a flow from non-participation to unemployment.

⁶Full sample construction, comparison to CPS flows, and regression specifications are provided in Appendix B

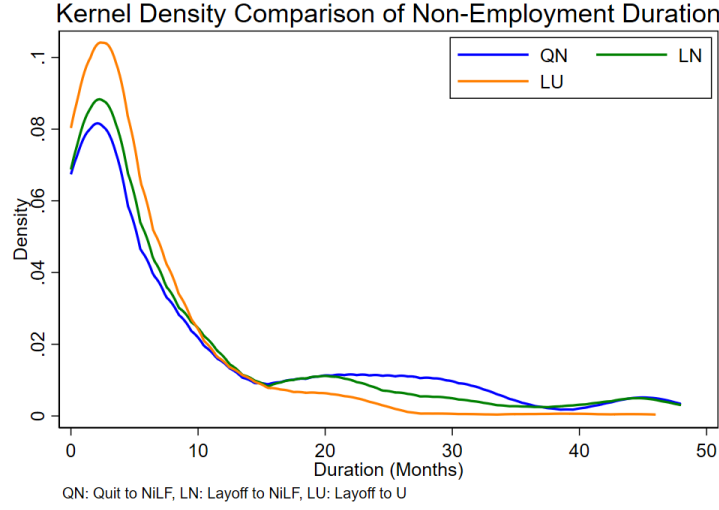


Figure 4: **Kernel density of non-employment duration by transition type (PSID): Quit to non-participation (QN), layoff to non-participation (LN), and layoff to unemployment (LU).**

Labor Market Frictions. Both non-participants and the unemployed must wait to receive a job offer before moving to employment. The offer arrival rate for the unemployed is $\lambda_u(Z)$ and, without loss of generality, is greater than the arrival rate for the non-participants, $\lambda_n(Z)$. Both rates are assumed to be higher in expansions than in recessions.

3.3. Value Functions

The state variable of an individual is $(a, z, p; \mathcal{S})$, where a is assets, z is their idiosyncratic productivity, p is whether they are exogenously constrained from participation, and $\mathcal{S}$ is the aggregate state, which includes the recession indicator Z . An unemployed worker will have an additional state for their UI eligibility, e .

We express an individual's decision problem recursively, using the following value functions. Let $V(\cdot)$ be the beginning of period value for an individual without an employment opportunity, $J(\cdot)$ be the value for an individual with an employment opportunity (ie: either they maintain employment from the last period or have a job offer), and $W(\cdot), U(\cdot), N(\cdot)$ be the values for an individual who has then chosen to be employed, unemployed, or a non-participant that period. For the employed,

$$W(a, z, p; \mathcal{S}) = \max_{c \geq 0, a' \geq 0} \log(c) + \beta E \left[\underbrace{(1 - \delta(Z))J(a', z', p', 0; \mathcal{S}')}_{\text{No Layoff}} + \underbrace{\delta(Z)V(a', z', p', 1; \mathcal{S}')}_{\text{Layoff}} \right]$$

$$\text{st} \quad c + a' = \max\{(1 + r(\mathcal{S}))a + (1 - \tau(\mathcal{S}))w(\mathcal{S})z, \underline{c}\}.$$

For the unemployed,

$$\begin{aligned}
U(a, z, p, e; \mathcal{S}) &= \max_{c \geq 0, a' \geq 0} \log(c) - \theta + \beta E \left[\underbrace{\lambda_u(Z) J(a', z', p', e'; \mathcal{S}')}_{\text{Job Offer}} + \underbrace{(1 - \lambda_u(Z)) V(a', z', p', e'; \mathcal{S}')}_{\text{No Job Offer}} \right] \\
\text{st} \quad & c + a' = \max\{(1 + r(\mathcal{S}))a + e \cdot b(z, \mathcal{S}), \underline{c}\} \\
& e' = e \quad \text{with probability } (1 - \mu(Z)); \quad = 0 \quad \text{o/w.}
\end{aligned}$$

For the non-participants,

$$\begin{aligned}
N(a, z, p; \mathcal{S}) &= \max_{c \geq 0, a' \geq 0} \log(c) + \beta E \left[\underbrace{\lambda_n(Z) J(a', z', p', 0; \mathcal{S}')}_{\text{Job Offer}} + \underbrace{(1 - \lambda_n(Z)) V(a', z', p', 0; \mathcal{S}')}_{\text{No Job Offer}} \right] \\
\text{st} \quad & c + a' = \max\{(1 + r(\mathcal{S}))a + h, \underline{c}\}.
\end{aligned}$$

The value for an individual without an employment opportunity depends on whether he or she is constrained to non-participation ($p = 0$) or not ($p = 1$). If the individual is not constrained, he or she can choose unemployment or non-participation freely each period:

$$V(a, z, p, e; \mathcal{S}) = \begin{cases} \max\{\underbrace{U(a, z, p, e; \mathcal{S})}_{\text{Unemployment}}, \underbrace{N(a, z, p; \mathcal{S})}_{\text{Non-participation}}\} & \text{if } p = 1 \\ N(a, z, p; \mathcal{S}) & \text{if } p = 0. \end{cases}$$

The value for an individual with an employment opportunity depends on whether or not he or she chooses to continue employment. The first line below features the arrival of a match continuation shock x . Notice that a worker who receives the continuation cost shock but doesn't pay is laid off and eligible for UI ($e = 1$), whereas the worker who quits is not eligible ($e = 0$):

$$\begin{aligned}
J(a, z, p, e; \mathcal{S}) &= \lambda_x(Z) \left[(1 - p) \max\{ \underbrace{W(a, z, p; \mathcal{S}) - x}_{\text{pay cost, keep employment}}, \underbrace{V(a, z, p, 1; \mathcal{S})}_{\text{layoff}} \} + p N(a, z, p; \mathcal{S}) \right] \\
&\quad \underbrace{\hspace{10em}}_{\text{Match Continuation Cost Shock}} \\
&+ (1 - \lambda_x(Z)) \left[(1 - p) \max\{ W(a, z, p; \mathcal{S}), \underbrace{V(a, z, p, 0; \mathcal{S})}_{\text{quit}} \} + p N(a, z, p; \mathcal{S}) \right].
\end{aligned}$$

3.4. Aggregate Shocks and Business-Cycle Dynamics

Aggregate shocks to job finding and job loss and TFP are the primary candidate drivers of business cycles in the model. We model the cyclical shocks as a one-time, unanticipated aggregate impulse in the sense of [Boppart et al. \(2018\)](#). Let z_t be a scalar recession state that follows

$$z_t = \rho z_{t-1} + \eta_t, \quad z_0 = 0, \quad \eta_t = \mathbf{1}\{t = 1\}; \quad (8)$$

that is, a single unit innovation hits at $t = 1$, and no further innovations are realized or anticipated, so $z_t = \rho^{t-1}$ for $t \geq 1$. Each cyclical parameter loads linearly on this common impulse, jumping to its bad-state value on impact and reverting geometrically to its good-state value,

$$\lambda_s^t = \lambda_s^* - \varepsilon_s z_t, \quad s \in \{u, n\}, \quad (9)$$

$$\delta^t = \delta^* + \varepsilon_\delta z_t, \quad (10)$$

where λ_s^*, δ^* are the good-state (steady-state) values and $\varepsilon_s = \lambda_s^* - \lambda_s^B$, $\varepsilon_\delta = \delta^B - \delta^*$ are the impact responses, so that $\lambda_s^1 = \lambda_s^B$ and $\delta^1 = \delta^B$. The same construction applies to the continuation-cost arrival λ_x , the UI-extension rate μ , and aggregate TFP.

Notably, selective layoffs may also vary over the business cycle. The arrival of a job continuation cost shock x is $\lambda_x(Z)$ and may be higher or lower along the recession than in the steady state. We will see later that the cyclicity of this parameter is only one determinant of whether the composition of layoffs moves towards higher or lower productivity workers in recessions.

Finally, to mimic the behavior of UI in the US in most business cycles since 1976, the probability of losing UI eligibility varies with the aggregate state, and we assume $\mu^G > \mu^B$ to reflect the longer duration of benefits during recessions.

4. Computation

We solve the model with the impulse response methods of [Boppart et al. \(2018\)](#). The period length is one month. The steady state is the stationary equilibrium of the economy held at the normal-times ($Z = G$) parameter values: we solve the value functions and decision rules on a grid over assets and productivity, find the labor tax rate that clears the government budget, and compute the stationary distribution over states.

A recession is a single MIT vector impulse. Every cyclical parameter $(A, \delta, \lambda_u, \lambda_n, \lambda_x, \mu)$ jumps to its bad-state value in the impact month, and thereafter, its deviation from the normal-times value decays geometrically at a common monthly rate ρ_x ; the economy is then computed forward to convergence. Along this perfect-foresight path we guess (i) the path of the capital-effective labor ratio, which determines prices, and (ii) the path of the labor tax that clears the government budget. We solve the household problem backward; simulate the distribution forward; and update the guesses until they converge. This technique is well-suited to our model because prices move little over the business cycle and the impact of tax rates on revenues is monotone within the range generated by all quantitatively relevant impulses we consider.

When we report a moment “at the recession trough,” we mean its largest absolute deviation from steady state along the impulse. Since the model has no growth trend, we compare the data’s peak-to-trough decline in real GDP per capita adjusted for trend growth over the same months.⁷

5. Estimation

We estimate the model by the simulated method of moments (SMM). The procedure is standard, and our contribution is additional moments. To the stocks and gross flows that discipline the standard model, we add the four series of Section 2: the level and the cyclicalities of quits, and the level and the cyclicalities of the share of layoffs flowing to non-participation. These four moments discipline the two ingredients we added to the workhorse model, random quits and selective layoffs, and how each moves over the cycle.

Our baseline estimation targets the era around the Great Recession, 2003–2019. We estimate in two stages. The 10 steady-state parameters (θ , h , δ^G , λ_u^G , λ_n^G , λ_x^G , ρ_z , p_0 , p_1 , and x) are estimated to match the normal-times averages (2002–2005) of 11 level moments: the employment-population ratio, the unemployment rate, the EU, EN, UE, NE, UN, and NU transition rates, the quit rate, the layoff rate, and the share of layoffs to non-participation. The five recession parameters (A^B , δ^B , λ_u^B , λ_n^B , λ_x^B) are estimated to match the *business-cycle volatilities* over the whole period. The volatility targets are the standard deviations of the HP-filtered logs of six series: the employment-population ratio, the unemployment rate, quits, layoffs, the layoff share to non-participation, and labor productivity (output per worker).⁸ UI does not expire during the recession ($\mu^B = 0$), matching the extensions to 99 weeks in the Great Recession, in normal times μ^G implies a mean benefit duration of six months.

The weighting matrix is judgmental. The employment-population ratio and the unemployment rate carry the heaviest weights, followed by our new targets: the quit rate, layoff rate, and layoff share to N.⁹ Since the empirical flows are not in steady state, a model that matched the exact EN, EU, UE, and NE flows would not match employment and unemployment, and so we weight these moments less and prioritize getting the change in the macro aggregates correct.

Several fixed parameters complete the calibration. The model period length is one month.

⁷Real GDP per capita fell 4.7% from 2007m12 to 2009m6; adding mean trend growth of 1.95% per year over those 18 months gives a 7.6% decline relative to trend, which we round to our 7.5% target. The same method gives 6.0% for the 1981–82 recession used in Section 11.

⁸The flow series are smoothed with a centered three-month moving average before filtering to remove month-to-month sampling noise.

⁹The weights on the level moments are 100 (E/pop), 100 (U rate), 10 (quits, layoffs, layoff share to N), and 5 on the rest, applied to squared proportional deviations of the normal-times levels.

The monthly interest rate delivers a 1% annualized return and $\beta = 1/(1 + r)$. The labor share is $1 - \alpha_y = 0.67$. The stationary variance of log productivity is 0.20.¹⁰ The UI system provides a replacement rate of $b_0 = 0.5$ for earnings up to the median wage, with the cap $\bar{b}$ binding thereafter. The consumption floor is $\underline{c} = 0.05$, approximately 5% of the median wage, capturing the Supplemental Nutrition Assistance Program, the primary welfare program utilized by temporarily unemployed workers (East and Simon (2024)).¹¹ The birth/death rate is $R_d = 0.5\%$ per month, which accounts for individuals leaving the survey without an observed separation and for new entrants who affect the NU, UE, and NE rates.

The Standard Model. Our comparison throughout is the same economy stripped of the two ingredients we have introduced. Selective layoffs are removed ($\lambda_x = 0$, $x = 0$), so all layoffs are random; and random quits are removed ($p_0 = 0$; $p_1 = 1$), so all quits are selective. It is estimated by the same procedure. However, it follows the standard approach in the literature and targets only flows across states (EU, UE, EN, NE) and not the flows by reason (quits, layoffs, and the share of layoffs to non-participation).

5.1. What the New Reason-Cross-Destination Moments Identify.

The inference rests on the simple economics of the model: where a separated worker goes reveals how much surplus the match had, and how many workers voluntarily separate reveals how many sit near the margin of participation.

Quits, LFPR, and Gross Participation Flows Inform Random Quits. In a standard model in which every quit is a marginal labor supply choice, the quit rate and the participation rate together put a hard restriction on the persistence of the productivity process driving quits. The persistence must be high enough to keep quitters out of the labor force long enough to match the steady-state participation level of 83%. However, this leads the standard model to deliver far too little re-entry from non-participation (NE + NU), relative to the data. A single productivity process struggles to generate high gross participation flows *and* keep 17% of the population out of the labor force. The participation-constraint shock adds a missing degree of freedom. It absorbs the quits that lead to persistent, or even permanent, exits (disability, family circumstances, a spouse's job), freeing the endogenous quits to better match gross flows. In summary, the labor force participation level and three participation flows (quits, NE, and NU) jointly influence the parameters of the participation-constraint shock (p_0, p_1): the LFPR determines how many quits the constraint must absorb;

¹⁰This variance accounts for censoring due to selective non-participation (Karahana and Ozkan (2013)).

¹¹East and Simon (2024) document that SNAP has the broadest eligibility among means-tested programs and shows the largest increase in receipt during unemployment, second only to UI.

and the gross flows adjust the margins determining the distribution of employment surplus, and movement within it.

The Cyclicalities of Quits Informs the Hoarding Margin. Quits fall in a recession only if the workers contemplating a quit care about a *temporary* collapse in job-finding rates. That sensitivity depends on the persistence of the shock pushing them toward the margin: a worker whose low surplus is near-permanent has little reason to hoard a job she will not want next year, while a worker whose surplus will drift back up wants to be employed when it does. With all quits endogenous, the persistence required to match the level of non-participation lowers this sensitivity. Splitting off the random quits relaxes the restriction on ρ_z , and the random-quit shocks are invariant at business-cycle frequency. The sensitivity of quits to the movements in the incentive to quit over the cycle closely informs how many quits are endogenous, marginal, elastic separations and how many are exogenous, detached, inelastic ones.

The Destination of Layoffs Inform Steady-State Selection in Layoffs. Matching the exit rate of 35.5% for laid-off workers requires that layoffs disproportionately target workers whose employment surplus is small enough that they prefer to leave the labor force. There are already incentives that tilt separated workers toward exit (the search cost and the forgone home production in unemployment), but quantitatively, these are not enough, and they cannot be freely chosen to match the share of exits after layoff because they also govern the incidence and duration of quits, the duration of non-participation after layoff, and the other attachment flows. The continuation-cost shock (λ_x) takes up the missing gap of exits that cannot be explained by these fundamentals by moving layoffs toward low-surplus workers who are more likely to exit.

The Cyclicalities of the Destination of Layoffs Informs Cyclical Selection. Part of the fall in the share of layoffs to non-participation during the recession is due to labor supply. Unemployment insurance extensions and a larger decline of the offer rate in non-participation, compared to unemployment, make searching relatively more attractive. Additionally, with endogenous layoffs, job hoarding plays a role in moving layoffs toward more attached workers just as it does with endogenous quits. But the observed fall is larger than these incentives can produce. The residual identifies a shift in the *selection* of layoffs up the surplus distribution: the movement of (δ^B, λ_x^B) relative to their normal-times values.

Parameter	Full model	Standard model
<i>Estimated</i>		
θ (disutility of active search)	0.095	0.097
h (home production level)	0.512	0.619
δ^G (random layoff, normal, %)	0.86	1.15
λ_u^G (offer rate, U, normal, %)	27.33	26.62
λ_n^G (offer rate, N, normal, %)	11.19	11.78
λ_x^G (cont.-cost shock, normal, %)	8.57	n/a
ρ_z (productivity persistence)	0.950	0.959
p_0 (remain participation-constrained)	0.960	0
p_1 (remain unconstrained)	0.998	1
x (continuation cost)	1.042	n/a
A^B (TFP in recession)	1.000	0.982
δ^B (random layoff, recession, %)	1.52	2.07
λ_u^B (offer rate, U, recession, %)	22.39	22.21
λ_n^B (offer rate, N, recession, %)	6.10	10.77
λ_x^B (cont.-cost shock, recession, %)	9.21	n/a
<i>Fixed</i>		
μ^G (UI expiry, normal, %)	10.91	10.91
μ^B (UI expiry, recession, %)	0.00	0.00
R_d (death/birth rate, %)	0.50	0.00
β (discount factor)	0.999	0.999
α_y (capital share)	0.330	0.330
b_0 (UI replacement rate)	0.500	0.500
$\bar{b}$ (UI cap)	1.000	1.000
$\underline{c}$ (consumption floor)	0.050	0.050
$\sigma_{\log z}^2$ (stationary variance)	0.200	0.200
<i>Equilibrium Budget Balance</i>		
τ^* (steady-state labor tax, %)	1.10	1.70

Table 5: **Model parameters** (monthly; probabilities in percent). Estimated parameters by SMM as described in the text. Standard errors and t-stats provided in the Appendix.

5.2. Parameter Estimates

Table 5 reports the estimates. Standard errors, a full Jacobian analysis, and sensitivity results are in Appendix D; the parameters are estimated precisely, with all t -statistics above 4.

The participation shock removes $1 - p_1 = 0.25\%$ of unconstrained individuals per month and holds them out of the labor force for an average of $1/(1 - p_0) = 25$ months; it generates about a third of all quits. The continuation-cost shock arrives at rate $\lambda_x^G = 8.6\%$ per month and the cost $x = 1.04$ is roughly one month of median earnings; only workers whose surplus is below that threshold (about 4% of the employed) choose a layoff when it arrives, generating

31% of layoffs.¹² The two non-market flow values are moderate: home production is $h = 0.51$, about two-fifths of mean labor earnings, and the disutility of search is $\theta = 0.09$, equivalent to a roughly 9% consumption tax while searching. These two are identified jointly by the employment-population ratio, the level of quits, and the destination of layoffs—the moments that locate the mass of workers near the participation margin.

Recessions in the full model are generated by flows, not TFP. Random layoff rates rise by three-quarters (0.86% to 1.52%), the arrival of continuation-cost shocks rises modestly (8.6% to 9.2%), the offer rate in unemployment falls by a sixth (27.3% to 22.4%), and the offer rate in non-participation falls by nearly half (11.2% to 6.1%). The sharp fall in λ_n relative to λ_u matters for attachment: in recessions, leaving the labor force becomes a much worse route back to work than searching, which is part of why more laid-off workers stay. The TFP shock is essentially zero ($A^B \approx 1$), and this is an outcome of the estimation rather than an assumption: layoffs in recessions target more highly productive workers than layoffs in the steady state. Consequently, effective labor falls more than the number of employed, so output per worker falls through composition. This channel is not operational in the standard model, which requires a decline in TFP ($A^B = 0.982$) to match GDP-per-worker volatility.

5.3. The Standard Model Does Not Produce the Composition of Separations or the Attachment Flows

Table 6 shows the fit of both models. The normal-times stocks and the business-cycle volatilities of employment, the unemployment rate, and output per worker are targeted and matched by both. Both also target the four standard gross flows directly (EN, NE, EU, UE), and so the new moments surrounding quits and layoffs distinguish the two models.

The Destination of Layoffs and the Level of Quits Pull in Opposite Directions.

In the standard model, 22.1% of laid-off workers exit the labor force, compared with 35.5% in the data. All layoffs are random, which means they hit average workers, and average workers are highly attached and search after displacement. The standard model must then manufacture the missing flows to non-participation as quits, and so requires a quits rate 50% larger than the one in data: 1.13% per month against 0.75%. No parameter can fix both at once. Raising h to push more workers toward the margin would send more laid-off workers to N, but it would also raise the quit rate, which is already too high. Matching the destination of layoffs *requires* that layoffs find the marginal workers, which is precisely what selective layoffs do: in the full model 31% of layoffs are selective, the share of layoffs to N is 34.6%, and the quit rate is 0.73%.

¹²Workers are forward looking and consider the total surplus, or present discounted values, of continuing work or not. It is not a comparison of current period flow values.

Moment	Data	Full model	Standard model
<i>Levels: normal times (2002–2005)</i>			
E/pop	79.4	78.7	78.3
Unemployment rate	3.40	3.39	3.38
EU rate	0.92	0.81	1.09
EN rate	1.48	1.16	1.44
UE rate	25.0	27.1	22.5
NE rate	7.0	6.4	7.2
Quit rate	0.75	0.73	1.13
Layoff rate	1.20	1.24	1.40
Share of layoffs to N	35.5	34.6	22.1
UN rate	18.0	10.9	10.2
NU rate	3.50	2.31	0.22
<i>Business-cycle volatility: std of HP-filtered log, 2003–2019 (%)</i>			
E/pop	0.91	1.03	0.91
Unemployment rate	11.0	14.3	11.0
Quit rate	10.1	8.6	1.80
Layoff rate	11.0	8.4	6.7
Share of layoffs to N	8.6	8.0	4.46
Labor productivity (GDP/worker)	0.71	0.50	0.71
<i>Untargeted</i>			
Mean short-N spell duration (months)	3.0	3.1	3.3

Table 6: **Model fit.** Moment sensitivity analysis provided in the Appendix.

Quits Move the Wrong Way over the Cycle. During recessions quits fall in the data but rise in the standard model. Because its layoffs are random and strike average-productivity workers, the standard model needs a fall in TFP to match output-per-worker volatility. This causes wages to decline, lowers the surplus of work, and incentivizes quits. And because every quit is endogenous, the productivity process must be persistent enough to deliver the level of non-participation. A more persistent process limits the job-hoarding response to a temporary decline in job-finding rates. In the full model TFP does not decline, the productivity process is less persistent, and hoarding dominates: quits fall, as in the data. The standard model likewise generates only half the observed volatility of exits after layoff, because the economic incentives encouraging countercyclical attachment, like UI extensions, are not strong enough on their own. This is conceptually how cyclical selection is inferred: it is the residual that cannot be explained by economic incentives.

Attachment Moves the Wrong Way. The standard model has almost no flows from non-participation into search: its NU rate is 0.2%, against 3.5% in the data. While [Krusell](#)

et al. (2017) and others solve this problem with exogenous idiosyncratic shocks to the cost of work, the full model generates two-thirds of these flows through fundamentals alone. It also reproduces their cyclicalities: during recessions NU rises to 5.4% (data: 5.6%) as non-employed workers near the margin respond to the collapse of λ_n by choosing active search, and UN declines through the same force and through UI extensions. The standard model does not reproduce rising attachment.

We emphasize that the failure of the standard model is not a simple calibration failure. It is being confronted with additional moments, and it is natural to expect it to need additional ingredients. But these moments are not superfluous. In the next three sections, we use the repaired model to reinterpret the business cycle: what shocks drive it (Section 6), and how those shocks are filtered through the two new mechanisms: selection in who separates (Section 7) and the labor supply of marginal workers (Section 8). We also provide more validation of these channels to external evidence not used in the calibration.

6. Sources of Business Cycles

We revisit the classic question of what drives recessions. Table 7 turns on one shock at a time (TFP, job finding $[\lambda_u, \lambda_n]$, job loss $[\delta, \lambda_x]$, and the UI extension) and reports the trough deviation of each moment.¹³ Agents understand and react to each counterfactual path.

Recessions Are Driven by Labor Market Frictions. The Great Recession is a product of labor-market shocks; TFP shocks play essentially no role. Contrary to Shimer (2012), the rise in unemployment in the full model is equally a job-loss and job-finding event. This is partially because increases in job loss in the full model come with a shift from selecting low surplus, low productivity workers towards random layoffs of average workers. This generates a larger increase in unemployment per laid-off worker. It also endogenously generates a decline in output per worker without requiring a fall in TFP. The standard model reads the same recession the other way around. It lacks a large increase in attachment in recessions, and so it leans on layoffs even more than a job-finding collapse to move unemployment. It lacks selective layoffs, and so it cannot generate output-per-worker volatility from composition and must assign a technology shock ($A^B = 0.98$) to carry the output-per-worker decline. Figure 5 plots the role of job-loss and job-finding as impulse responses. Other episodes are examined in Section 11.

Unemployment Insurance Extensions. UI extensions alone deliver a 0.28 point rise in unemployment (about a ninth of the total increase), a result we examine in Section 9. The

¹³Appendix Table 21 reports the same decomposition in volatility terms, as the standard deviation each shock generates on its own.

Moment	All shocks	TFP	Job finding	Job loss	UI extension
<i>Full model</i>					
Δ E/pop	-2.94	-0.01	-1.31	-1.56	-0.13
Δ U rate	+2.50	-0.03	+1.15	+1.24	+0.28
Δ EU rate	+0.63	-0.00	-0.03	+0.60	+0.05
Δ UE rate	-4.90	-0.00	-4.90	-0.00	-0.00
Δ EN rate	-0.49	+0.00	-0.55	+0.11	+0.01
Δ NE rate	-2.62	+0.01	-2.38	+0.16	-0.07
Δ UN rate	-5.93	-0.31	+4.51	-1.20	-3.32
Δ NU rate	+3.04	-0.08	+3.90	-0.18	-0.12
Δ Quit rate	-0.46	+0.00	-0.46	-0.01	-0.00
Δ Layoff rate	+0.60	+0.00	-0.10	+0.72	+0.05
Δ Share of layoffs to N	-12.97	+0.00	-4.63	-6.64	-1.30
GDP decline	2.56	0.03	1.18	1.13	0.10
<i>Standard model</i>					
Δ E/pop	-2.71	-0.78	-0.22	-1.49	+0.18
Δ U rate	+1.79	-0.19	+0.36	+0.91	+0.54
Δ EU rate	+0.47	-0.05	+0.06	+0.40	+0.02
Δ UE rate	-0.66	-0.00	-0.66	-0.00	-0.00
Δ EN rate	+0.36	+0.40	-0.36	+0.21	-0.04
Δ NE rate	-1.52	-0.07	-1.21	+0.12	-0.15
Δ UN rate	-6.73	+3.63	-2.98	+0.42	-5.95
Δ NU rate	+0.30	-0.14	+0.85	-0.03	-0.11
Δ Quit rate	+0.31	+0.34	-0.31	+0.09	+0.04
Δ Layoff rate	+0.52	-0.00	-0.00	+0.52	-0.00
Δ Share of layoffs to N	-3.77	+3.67	-4.16	+0.38	-1.72
GDP decline	5.94	5.29	0.38	1.36	0.04

Table 7: What drives recessions? Trough deviation of each moment from steady state (percentage points) along the recession impulse when only one shock is turned on at a time. “Job finding” moves (λ_u, λ_n) ; “Job loss” moves (δ, λ_x) ; “UI extension” sets $\mu^B = 0$.

attachment flows show how policy and frictions interact. Job finding acting alone would push the UN rate up 4.9 points: with offers scarce and benefits expiring on schedule, unemployed workers near the margin give up and exit. The UI extension acting alone pushes UN *down* 2.9 points, holding the displaced in the labor force. In the full recession, the extension wins: UN falls by 3.3 points, as in the data. Measured unemployment in a US recession is partially a product of policy-supported attachment in both the full and standard models.

The shock decomposition tells us which exogenous forces hit the economy. In the next two sections, we explore how agents in the economy respond and how this response affects the transmission of shocks.

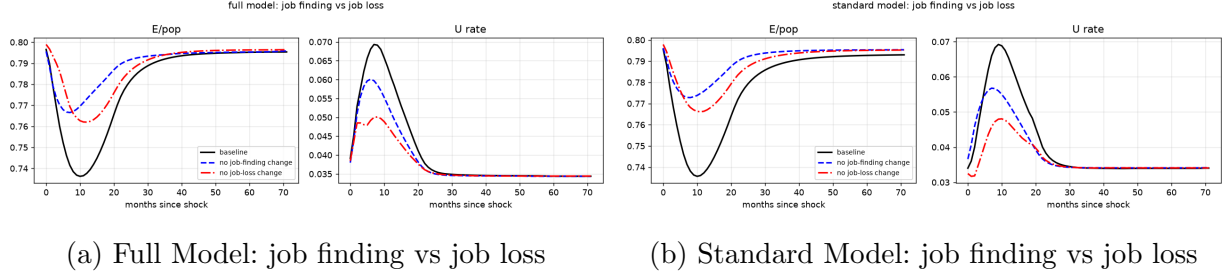


Figure 5: **Role of job-finding and job-loss shocks in business cycle dynamics.** Baseline impulse response versus counterfactuals holding the indicated shock at its normal-times value.

7. Selection, Labor Supply, and the Marginal Workforce

The Marginal Workforce. Marginal workers are those whose net employment surplus is near zero. They are the workers choosing selective quits and layoffs, which makes their labor supply choices the key elastic margin of the economy. The model identifies them sharply: a worker is at risk of a selective quit if her surplus is negative, and at risk of a selective layoff if her surplus is positive but below the continuation cost x . The likelihood of either falls with idiosyncratic productivity and rises with assets. For low-productivity individuals, home production and UI (if laid off) replace a large share of potential labor income, while high-asset individuals can use their savings to smooth consumption through non-employment. The destination of a separation is therefore a signal of where a worker sits relative to this margin: workers who quit (selectively) to non-participation have the lowest pre-separation residual earnings and those laid off (randomly) into unemployment the highest. Non-employment spells are the shortest for layoffs to unemployment and longest for flows to non-participation. These are exactly the orderings that match the PSID (Figures 3 and 4). The finer flow classification, distinguishing destination as well as reason, carries information about distance to the margin that a coarser quit/layoff split did not.

Table 8 reports the distribution. In normal times, the marginal workforce (negative surplus plus negative-if-cost) is 8.7% of the prime-age population, and another 5.2% are detached by the participation constraint. It is concentrated exactly where the data suggested: 30.8% of non-participants are marginal and a further 28.0% constrained, against 3.7% of the employed and 2.7% of the unemployed. The recession *shrinks* the marginal workforce everywhere (among non-participants, the negative-surplus group falls to 26.9%). Scarcity raises the value of holding a job and increases the surplus everywhere. By contrast, the share of the population with positive surplus actually falls in recessions in the standard model because of the decline in TFP.

Figure 6 plots the surplus distribution over the unconstrained population in both models,

Category	Total	Emp.	Unemp.	NILF
<i>Full model</i>				
Participation shock	5.2, 5.2	0.0, 0.0	0.0, 0.0	28.0, 26.3
Negative surplus	3.3, 2.7	0.0, 0.0	0.0, 0.0	17.6, 13.7
Negative surplus if continuation cost	5.4, 5.2	3.7, 3.4	2.7, 0.6	13.2, 13.2
Positive surplus always	86.1, 86.9	96.3, 96.6	98.3, 99.4	41.2, 46.8
<i>Standard model</i>				
Participation shock	0.0, 0.0	0.0, 0.0	0.0, 0.0	0.0, 0.0
Negative surplus	7.6, 7.9	0.0, 0.0	0.0, 0.0	39.8, 38.6
Negative surplus if continuation cost	0.0, 0.0	0.0, 0.0	0.0, 0.0	0.0, 0.0
Positive surplus always	92.4, 92.1	100.0, 100.0	100.0, 100.0	60.2, 61.4

Table 8: **Population by employment-surplus category (percent): normal times, recession trough.** The marginal workforce comprises the two middle (blue) categories: workers with negative surplus (they quit if employed) and workers whose surplus is positive but below the continuation cost x (they are released by a selective layoff if the shock arrives).

in normal times and at the trough, with the quit region (< 0) and the selective-layoff region (0 to x) shaded. The full model places 9.1% of the unconstrained population in the two shaded regions, and in recessions, the distribution shifts right, and the regions thin out. The standard model has no selective-layoff region at all, and instead of shifting in a recession, the distribution of net surplus increases in variance.

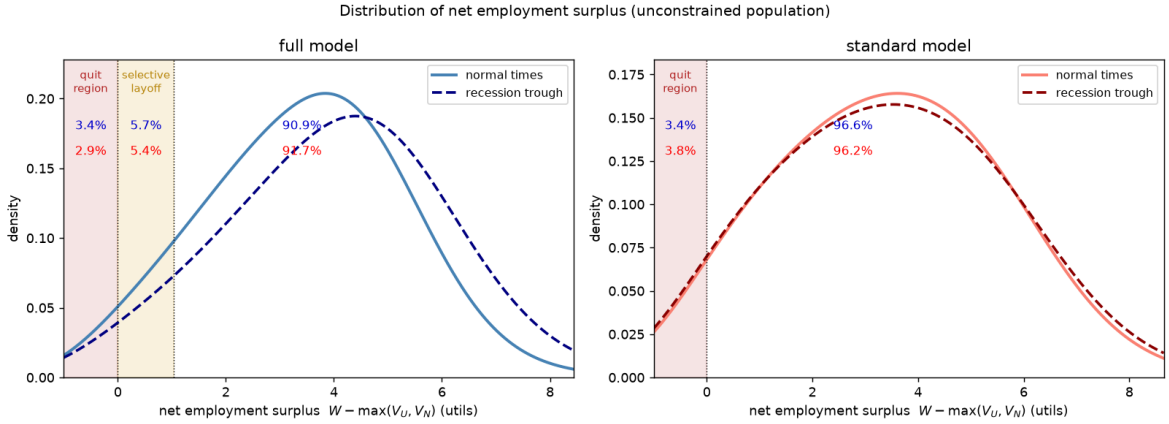


Figure 6: **Distribution of net employment surplus** ($W - \max(V_U, V_N)$, surplus relative to quitting) over the unconstrained population. Shaded: the quit region (< 0) and the selective-layoff region (0 to x). Region masses in blue (normal times) and red (recession trough).

Share	Full model		Standard model	
	Normal	Recession	Normal	Recession
Share of quits selective	66.2	19.0	100.0	100.0
Share of layoffs selective	30.6	17.7	0.00	0.00
Share of U stock entered selectively	3.28	2.01	0.15	0.39
Share of N stock entered selectively	44.9	42.2	73.5	74.8

Table 9: **Role of Selection: stocks and flows.** Selective quits are endogenous (negative-surplus) quits; selective layoffs are continuation-cost (x -shock) layoffs; the remainder are exogenous. A non-employed worker is counted as having entered selectively if the current non-employment spell began with a selective quit or layoff. Recession values are taken at the point of largest deviation during the episode.

7.1. How Selection Moves over the Cycle

Table 9 reports the selective share of each separation flow and of each non-employment stock. In normal times in the full model, 66% of quits and 31% of layoffs are selective, initiated by, or aimed at, low-surplus workers. Both shares fall sharply in recessions. The selective share of quits collapses to 19%, as marginal workers hoard their jobs and the quits that remain are dominated by the random quits of constrained workers. The selective share of layoffs falls to 18%, as the recession’s separations are increasingly random and hit attached workers. Whether selection should rise or fall in recessions is obvious neither a priori nor from inspection of the parameters alone. The continuation-cost shock arrives *more* often in recessions, with λ_x rising from 8.6% to 9.2%, but more workers are willing to pay the continuation cost because scarcity has raised the value of every job. The estimation finds the right mix to reproduce the joint fall in quits and in the share of layoffs to N. The logic mirrors that of the identification argument of Section 5.1: since quits fall in recessions, the net surplus of marginal workers must be rising, and we should see *more* laid-off workers stay in the labor force; but the observed fall in the layoff share to N is larger than labor supply alone can deliver, so the incidence of layoffs must be shifting toward attached workers.

Some of the workers who quit endogenously make a later NU transition when their productivity drifts upwards, but this is small. What is more common is for laid-off workers to make a UN transition when they exhaust their UI, and this contributes to the roughly one-quarter of unconstrained non-participants that did not enter via selective quit.

Validation of Cyclical Selection in the Data. Should we believe the model’s estimate of the cyclical de-selection of layoffs? We bring in external evidence from [Mueller \(2017\)](#) for a validation test. Using CPS data, Mueller documents that the pool of workers separating into unemployment shifts strongly toward high-productivity workers in recessions. This is a fact our estimation never targets; we target only flows and steady-state statistics of the aggregate

Dependent variable	Full model	Standard model	Mueller (2017)
<i>Individual Productivity of Separated Workers</i>			
z (unemployed)	2.60	-0.65	2.77
z (layoff)	2.09	0.20	
z (quit)	1.74	0.43	

Table 10: Cyclical composition: coefficients on the unemployment rate along the recession impulse path, in the spirit of [Mueller \(2017\)](#).

wage distribution. Table 10 shows the results from running the same test [Mueller \(2017\)](#) ran on CPS data on the data generated by our model. Specifically, we regress the productivity of separators on the unemployment rate along the impulse path. The full model delivers a coefficient of 2.6 for workers transitioning from employment to unemployment, close to Mueller’s empirical estimate of 2.77, despite the two approaches being fundamentally different. The standard model delivers the opposite sign. While Mueller did not distinguish between quits and layoffs, the model shows both of these shifting toward more productive workers as both become more random, with the composition of layoffs driving the composition of EU.

The Role of Selection in Business Cycle Dynamics To quantify how selection and labor supply affect the transmission of business cycle shocks, we construct one counterfactual for each. Table 11 reports both as steady-state to trough changes, and Appendix Table 22 reports them in volatility terms. The *selection role* compares the baseline to a recession in which the arrival rate of the continuation cost is re-solved so that the layoff share to N stays at its steady-state value while the total layoff rate path is unchanged. The *labor supply role*, taken up in Section 8, compares the baseline to a counterfactual in which all decision rules are held at their steady-state values except the selective layoff choice which is chosen to generate the same split of layoffs as the factual model. This isolates how behavioral changes in quits, search, and participation contribute to the cycle.

Cyclical selection buffers the employment decline by 0.72 points at the trough and amplifies the unemployment rise by 0.48. In the baseline, recessions shift layoffs toward attached workers, and attached workers search after displacement. In the selection-neutral counterfactual, the same share of layoffs keeps falling on marginal workers who exit. Selection also does most of the compositional work: without it, the share of layoffs to N would fall only about a point in the recession, against 13 points in the baseline. This decline is driven by longer UI and relatively higher job finding rates in U than N. Figure 7 plots the selection-neutral impulse response.

Moment	Full model				Standard model	
	Steady state	Baseline Δ	Labor supply role	Selection role	Baseline Δ	Labor supply role
E/pop	78.7	-2.94	+1.00	+0.72	-2.71	-0.44
U rate	3.39	+2.50	+0.32	+0.48	+1.79	+0.19
UN rate	10.9	-5.93	-2.24	-2.67	-6.73	-1.25
NU rate	2.31	+3.04	+3.14	+2.31	+0.30	+0.40
Quit rate	0.73	-0.46	-0.42	-0.30	+0.31	+0.35
Layoff rate	1.24	+0.60	+0.01	-0.14	+0.52	+0.00
Share of layoffs to N	34.6	-12.97	-3.23	-11.98	-3.77	-3.29
Want-to-work share	18.1	+3.53	-0.41	-0.42	+2.39	+0.12
GDP decline	—	2.56	-0.96	-0.10	5.94	+0.24

Table 11: **Roles of labor-supply response and selection in layoffs (percentage points).** Entries are trough deviations from steady state. “Labor supply role” is a counterfactual with decision rules at steady state and the baseline incidence of selective layoffs fed in; the positive entry for E/pop means the labor-supply response buffers the employment decline. “Selection role” is baseline minus a recession re-solved so the layoff share to N stays at steady state while matching the baseline layoff-rate increase.

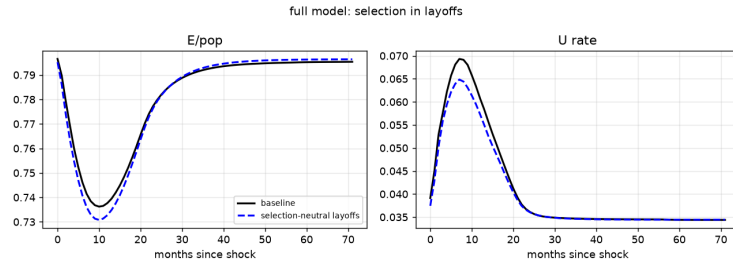


Figure 7: **Selection in layoffs.** Baseline impulse response versus the selection-neutral recession (layoff share to N held at steady state, layoff rate path matched).

8. The Labor Supply Channel in the Recession

In the full model, the labor supply response makes employment fall *less*, by 1.0 percentage point at the trough, and makes the unemployment rate rise *more*, by 0.32 points (Table 11). In other words, it buffers the employment decline by about a quarter and increases the unemployment rise by 13%. Interestingly, these happen to be of the same orders of magnitude as those of the selection role in Section 7.1, and in the same direction. Two factors drive this counter-cyclical labor supply. The first is job hoarding: the 0.46 point decline in the quit rate is via marginal workers who choose to remain employed but who would have quit at steady-state policies. The second is added workers: the labor supply response of non-participants more than accounts for the rise in the NU rate (a role of +3.14 against the +3.04 total rise), as the net surplus of working rises and jobs are relatively easier to find in U. Figure 8 plots

the labor supply role in the impulse response paths.

The standard model’s labor supply has the opposite impact on employment. Its response *deepens* the employment decline (by 0.44 points) as quits *rise* in the recession. With no selective layoffs, and a productivity-driven recession that turns marginal workers’ surplus negative, the only labor supply margin it has is the discouragement of those marginal workers, who respond to the downturn by exiting. In this sense, the employment-stabilizing impacts of labor supply choices are absent from the standard workhorse model.

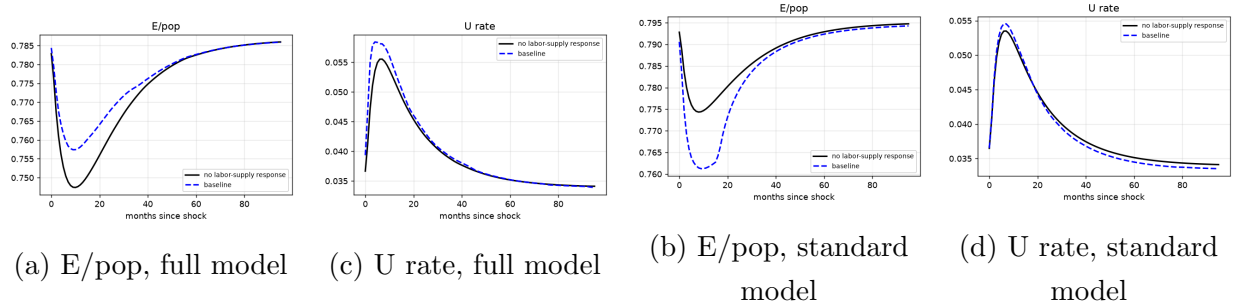


Figure 8: **Role of labor supply in recessionary dynamics.** In each panel, the dashed (baseline) line is the model’s response, and the solid line is the counterfactual with the labor-supply response shut off (decision rules frozen at steady state, baseline incidence of selective layoffs); the gap between them is the labor-supply role.

Why Marginal Workers Stop Quitting The recessionary decline in quits could bundle two forces: marginal workers may hoard because job finding has declined, or because rising job loss rates and longer UI benefits have made waiting for a layoff more attractive. The shock decomposition of Table 7 shows the job-finding column accounts for the entire quit decline. Re-solving the recession holding the job-finding path (λ_u, λ_n) at its steady-state level confirms this. It not only eliminates hoarding but reverses it: the labor-supply quit response flips from -0.28 to $+0.29$ points, and quits rise. Holding the recession UI extension (μ) at its steady-state level leaves the quit response essentially unchanged. When jobs are scarce, marginal workers quit less, because they know it will be hard to find a job when their productivity drifts upwards in the future. The same hoarding response exists in the standard model but is entirely offset by the decline in TFP, which makes working less attractive and raises the incentive to quit.

Dynamics of Marginal Workers over the Recession. To trace these labor supply dynamics at the worker level, we tag cohorts at the onset by their pre-shock employment surplus and follow their employment over the impulse response. Figure 9 splits the employed

into the bottom and top quartiles of net surplus $W - \max(U_0, N)$. The marginally attached employed hoard: their quit hazard falls, causing their employment to increase briefly, but over time, the rise in layoffs and the decline in job finding dominate and lower their employment rate. The core employed never quit in the steady state and so have no offsetting job-hoarding margin. They immediately lose employment through higher job loss and lower job finding, but eventually recover more quickly than the marginally attached employed because they are more likely to stay in the labor force and search in unemployment.

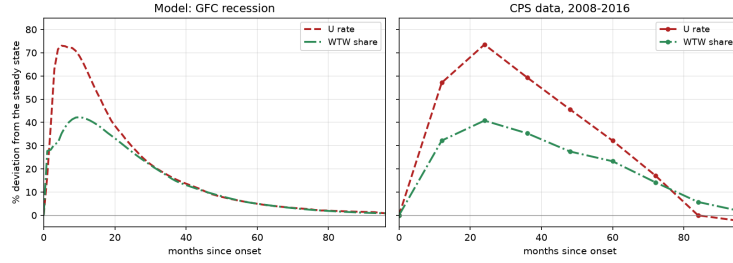


Figure 9: **Marginal-worker event study.** Change in employment rate of cohorts tagged at the recession onset and followed through the baseline impulse.

8.1. Validation: The Want-to-Work Share in the CPS

The model estimates that labor supply is counter-cyclical: everyone should be more willing to work, and this pushes more non-employed workers over the margin of having a higher surplus from employment than their current state. We call these individuals the “jobless who want to work” (WTW); specifically, they would accept a job if offered $W > \max\{U, N\}$. This is different from the concept of unemployment since the unemployment designation requires active search. Fortunately, the CPS contains an empirical counterpart of our WTW concept: the unemployed plus non-participants who answer “yes” to the question of whether they want a full-time or part-time job.¹⁴ Figure 10 compares the two over the GFC: the model predicts WTW is roughly a third as cyclical as the unemployment rate, and the CPS series moves by the same proportion. We take this as validation that the magnitudes of the labor supply response in our model are realistic, and we return to what this means for the practical issue of tracking slack in Section 10.

¹⁴The want-a-job question is available only in the ASEC file and thus at an annual frequency, which is why the CPS series in Figure 10 is less smooth.



(a) Full Model

Figure 10: **Unemployment rate versus the want-to-work share over the recession.** Deviation as percent of the steady state along the impulse response. Left: Full model. Right: CPS data 2008-2016.

Summary of the Two Channels. Together, the two new channels mitigated the fall in employment in the GFC by 1.16 percentage points and added 0.74 points to peak unemployment.¹⁵ These facts work together as stabilization forces as part of the unemployment spike is stabilization via workers staying attached and searching rather than leaving. We caution that these numbers are episode-specific and depend on trends in factors influencing labor supply as much as the specific shock. We illustrate this with additional episodes in Section 11.

9. The Welfare Cost of Business Cycles and the Value of UI

The previous sections established that (i) recessions displace workers less selectively, and (ii) marginal workers become more attached in recessions. Both change the welfare cost of a recession, through two distinct channels: composition replaces TFP as the source of the output decline, so no economy-wide productivity loss is needed, and the layoffs that do occur in normal times are concentrated on the workers with the least surplus to lose. To explore this concept, we compute, for every agent at the steady state, the permanent consumption-equivalent (CE) change that makes her indifferent between remaining at the steady state and entering the recession episode at date zero. Table 12 reports the results.

The mean welfare cost of the recession episode in the full model is 0.63% of lifetime consumption, against 1.55% in the standard model. The smaller TFP decline accounts for 40% of the gap between the models, and selection for 60%. By labor market state at the onset, non-participants lose 0.99%, the unemployed 0.83%, and the employed 0.54%. That non-participants lose more than the employed is a direct consequence of the net value of employment moving primarily through declines in the job finding rate. The employed lose more in the standard model because of the negative TFP shock that calibration requires.

¹⁵These numbers are from an exercise where both channels are turned off simultaneously, and interaction causes the combined impact to be less than the sum of their individual impacts.

	Cost of recession		Value of UI extension	
	Full	Standard	Full	Standard
Mean	0.63	1.55	-0.120	-0.005
Std. dev.	0.24	0.15	0.052	0.045
<i>Mean by labor market state at onset</i>				
Employed	0.54	1.58	-0.126	-0.012
Unemployed	0.83	1.37	0.095	0.221
Non-participant	0.99	1.45	-0.123	-0.009
Cost with selection-neutral layoffs (full)	1.00	–	–	–

Table 12: **Welfare costs of recessions in consumption equivalents (%)**.

Selection in layoffs makes recessions less costly. We re-run the calibrated full model but change layoffs to be selection-neutral: the same aggregate layoff path falling randomly across workers rather than concentrating on those with the least to lose. This raises the mean cost of recessions from 0.63% to 1.00% and highlights that selected workers have lower costs of displacement. They have smaller employment surpluses to destroy, and their fallback options (home production, and UI if laid off) replace a larger share of what they lose. To be clear, it is not the *change* in selection over the cycle that is driving this result. It is low-surplus workers’ higher layoff risk throughout the cycle.

The standard model overstates the cost of recessions primarily because it requires a fall in TFP that hurts *every* worker in the economy. Workers are paid less, and jobs are less attractive to the non-employed. This contrasts with our full model, where the only non-marginal workers who are affected are those who actually experience job loss and there is an increase in everyone’s UI tax rate. This is also why the cost of recessions is flatter across states in the standard model. For distributional questions, and for any policy aimed at the displaced or the marginally attached, the models give different answers.

UI Extensions. Our baseline recession includes the extension of UI benefits, as in the GFC. Solving the same recession with benefits expiring at their normal-times rate isolates the extension’s value. The unemployed at the onset gain 0.095% of lifetime consumption. The extension also contributes to attachment dynamics by decreasing UN flows and attracting laid-off workers to search. But the money for UI comes from the employed and non-participants, who each lose about 0.12% of lifetime consumption through higher taxes. As shown in Section 6, the extension raises the unemployment rate by 0.3 percentage points and also increases the layoff rate as marginal workers are less likely to pay the match continuation shocks if their outside option of UI is better. The population-weighted result is -0.12% of lifetime consumption, meaning that the average agent would pay to remove the

extension.

The standard model scores the same policy as approximately neutral (-0.005%). This is largely because layoffs target more attached workers and so unemployment barely responds to the policy. We temper this reading with the caveat that our experiment holds job creation fixed, omitting the vacancy response studied in [Hagedorn et al. \(2015\)](#) and the aggregate-demand effects studied elsewhere. Nonetheless, we think our framework is helpful because what the elasticity of the labor supply margin and selection into layoffs are important for any evaluation of UI.

10. Tracking Slack: Unemployment Versus Want-to-Work

If unemployment conflates displacement, labor supply, and policy, what should we track instead? The model offers a natural answer: the want-to-work stock of Section 8.1, the non-employed whose value of employment exceeds the value of their current state regardless of whether they are unemployed. In this section, we review how WTW relates to unemployment.

First, WTW is larger than unemployment. In the steady state, 12% of the population is not employed, is not participation constrained, and has a positive net surplus of employment, while the unemployment-to-population ratio is 2.9%. Unemployment does not capture idle labor supply, because there are non-participants with positive employment surplus who choose to wait on the slower offer rate λ_n rather than pay the cost of search and losing home production. Second, WTW is a third as cyclical (Figure 10) and so comparable to the CPS counterpart. The unemployment rate overstates the *proportional* deterioration in job-wanting and understates its breadth. Third, and most important for measurement, the wedge between the two series is partly labor supply and partly selection. Both the UI extension and the reduced selectivity of layoffs send laid-off workers to U rather than N in recessions (as in the data), and the rise in NU (as in the data) draws workers with positive surplus into measured unemployment. Both flows raise unemployment by more than they raise WTW, because they move workers who already wanted to work from N into U. The UI extension alone moves the unemployment rate by 0.3 percentage points while moving WTW by a tenth of a point. The unemployment rate is a measure of who is *searching*, and search responds to benefits; WTW is a measure of who would *work*, and that is the object that matters for slack.

The practical implication is that the unemployment rate is most distorted in recessions with active UI policy. A rise in unemployment during an extension episode mixes displaced workers, marginal workers drawn into measured search by benefits, and laid-off workers who would have exited had there not been an extension. We recommend more attention be paid during such episodes to the CPS WTW series we constructed, which can be easily constructed

by anyone using IPUMS. The consistency of the CPS want-a-job responses with the model’s prediction (Section 8.1) suggests that tracking this question could be an additional useful indicator for real-time measurements of slack.

11. Episodes: 1980s, Great Recession, and 2020s

The baseline estimation disciplined the model with a single recession. The distribution of net employment surplus, however, is always changing over time. For example, [Ellieroth and Michaud \(2024\)](#) argue that changes in women’s labor force attachment over the late 20th century fundamentally changed the response of job hoarding to the cycle. This means the labor supply backbone of our model is not time-invariant and needs to be re-calibrated to account for structural changes in participation elasticities.

We re-estimate the model on two further episodes with targets taken from CPS windows disciplining the steady-state to be contemporary to the episodes. First, we consider the 1980s recessions by re-estimating the full model with targets computed during the 1976-1989 period. UI extensions are capped at a median duration of nine months (roughly 39 weeks), reflecting the less generous Federal Supplemental Compensation system of that era. For the 2020s, we consider the 2022-2025 period. The latter half of this episode is considered the steady state, and the roughly 2022-23 period provides a “booming” labor market. UI duration is held at 6 months over this period.

Table 13 reports the parameter estimates and fit for each episode. One important thing to notice is that it shows the range the model we developed can achieve. The fact that TFP is estimated to be held constant in the GFC is not a tautology: the model requires a decline in TFP to generate the GDP-per-worker decline in the 1980s recessions. Table 14 decomposes each episode into the contribution of each shock and into the labor supply and selection roles; Figure 11 plots the impulse responses.

The 1980s Recessions. The 1980s recessions feature a larger role for selection and smaller role for labor supply. Job loss becomes a driver because we include with it the change in selection as job loss rates increase: it delivers 2.5 points of the decline in employment as the random layoff rate more than doubles (0.78% to 1.93%). While the drop in TFP is 3.1 percentage points, its decline contributes only 0.65 points to the employment decline. The era’s less generous UI also changes what the unemployment rate measures as the impact of UI policy becomes negligible. The rise in unemployment in the 1980s was overwhelmingly due to displaced workers searching, not benefit-supported attachment, yet attachment still rose, as in the data: the model reproduces the UN decline (13.2% to 11.2%, versus 19.6% to 15.4% in the data) and the NU rise (3.0% to 3.9%, versus 2.8% to 3.9% in the data) of that

	GFC (baseline)	1980s	2022–23 boom
<i>Parameters: steady state, episode state</i>			
θ (disutility of active search)	0.095	0.093	0.090
h (home production level)	0.512	0.504	0.509
δ^G (random layoff, %)	0.86	0.78	0.85
λ_u^G (offer rate, U, %)	27.33	28.38	29.08
λ_n^G (offer rate, N, %)	11.19	10.49	10.63
λ_x^G (cont.-cost shock, %)	8.57	8.41	7.61
x (continuation cost)	1.042	1.087	1.082
A^B (TFP)	1.000	0.969	1.007
δ^B (random layoff, %)	1.52	1.93	0.76
λ_u^B (offer rate, U, %)	22.39	21.63	29.08
λ_n^B (offer rate, N, %)	6.10	5.36	13.26
λ_x^B (cont.-cost shock, %)	9.21	9.97	5.33
UI in episode	no expiry	39 weeks	26 weeks
<i>Model outcomes: normal $\rightarrow$ episode</i>			
E/pop	78.7 $\rightarrow$ 75.7	77.2 $\rightarrow$ 72.6	80.0 $\rightarrow$ 81.0
U rate	3.39 $\rightarrow$ 5.9	3.51 $\rightarrow$ 6.5	3.44 $\rightarrow$ 3.03
Share of layoffs to N	34.6 $\rightarrow$ 21.6	36.8 $\rightarrow$ 22.4	33.5 $\rightarrow$ 31.1
Share of layoffs selective	30.6 $\rightarrow$ 17.7	35.2 $\rightarrow$ 17.9	30.7 $\rightarrow$ 26.2
UN rate	10.9 $\rightarrow$ 5.0	13.2 $\rightarrow$ 11.2	11.7 $\rightarrow$ 15.0
NU rate	2.31 $\rightarrow$ 5.4	2.96 $\rightarrow$ 3.89	3.08 $\rightarrow$ 2.15
Want-to-work share	18.1 $\rightarrow$ 21.6	20.1 $\rightarrow$ 24.8	17.2 $\rightarrow$ 16.0

Table 13: **Episodes: Calibration and fit.**

	GFC (baseline)	1980s	2022–23 boom
<i>Contribution to episode Δ U rate (acting alone, pp)</i>			
Job loss	+1.24	+2.04	+0.25
Job finding	+1.15	+1.25	-0.61
TFP	-0.03	+0.04	+0.01
<i>Contribution to episode Δ E/pop (acting alone, pp)</i>			
Job loss	-1.56	-2.54	+0.70
Job finding	-1.31	-1.04	+0.16
TFP	-0.01	-0.65	+0.11
<i>Impact of labor supply and selection (pp at trough)</i>			
Labor supply, E/pop	+1.00	+0.20	-0.29
Selection, E/pop	+0.72	+1.05	+0.27
Labor supply, U rate	+0.50	+0.21	-0.16
Selection, U rate	+0.55	+0.78	-0.17

Table 14: **Episodes: Recession decomposition.** The top panels give each shock’s contribution, acting alone, to the episode change in the unemployment rate and E/pop. The bottom panel gives the impact of the labor-supply response and of cyclical selection at the trough (boom peak for the 2020s), each measured as the baseline outcome minus the counterfactual with that channel shut off; a positive E/pop entry means the channel holds employment up.

episode, driven by job scarcity rather than policy.

The 1980s recessions are more impacted by the change in selection in part because the steady state from the 1970’s through the 1980’s features more selection to start with: 37% of normal-times layoffs flow to non-participation (the CPS target is 41%). The recession undoes this feature: the selective share of layoffs collapses as random layoffs take over, and the share of layoffs to N falls to 22% (target 29%). In the model, the cyclical “upgrading” of the separating pool documented by [Mueller \(2017\)](#) is in full force in the model’s 1980s.

The 2020s Expansion. The model interprets the 2020s expansion as held back by a significant decline in labor supply. For context, the decline in labor supply held back E/pop growth by 0.3 percentage points, which is about a third of the actual increase during these years. Figure 11 shows that the decline in unemployment considerably understates the decline in idle labor supply: the share of non-employed who wanted a job fell by much more. This is exactly the divergence between slack and measured unemployment that Section 10 emphasizes, now running in the expansionary direction.¹⁶

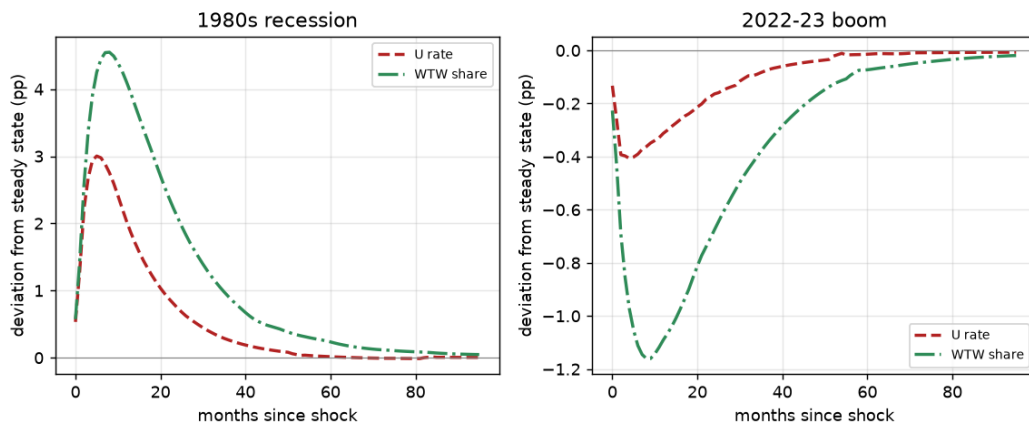


Figure 11: **Episodes.** Impulse responses of the unemployment (U) rate and the non-employed who want-to-work (WTW) share as deviations from each episode’s own steady state: 1980s recession, and 2022–23 boom.

Figures 12 and 13 visualize the labor-supply and selection decomposition of each episode. In every episode, the dashed (no labor supply) counterfactuals sit outside the baseline for employment and inside it for unemployment, implying qualitatively the same stabilizing pattern as in the GFC. Yet, each episode is distinct as labor supply plays a much smaller

¹⁶The job-to-job quit surge of the Great Resignation is a separate phenomenon: in our model, the cyclical force on quits is quits of marginal workers to non-employment, whereas the 2021–22 quits researchers studied were without non-employment spells ([Bagga et al. \(2023\)](#), [Cai and Heathcote \(2023\)](#), [Şahin and Tasci \(2022\)](#), [Michaels \(2024\)](#)).

role in the 1980s recessions. The dash-dotted (no selection) line follows the same stabilizing pattern for the GFC and the 1980s, but not for employment in the 2022-23 boom. Selection moved toward more random layoffs in 2022-23, a pattern we typically saw in recessions. This may suggest that the relationship between labor market tightness and selectivity in layoffs is non-monotone, or that the 2020s were a special episode for a variety of reasons.

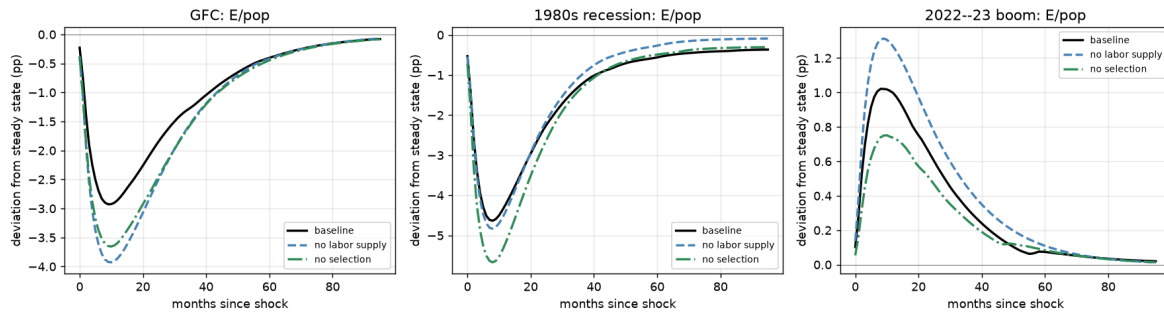


Figure 12: **Labor supply and selection in employment, by episode.** E/pop response of each episode (deviation from its own steady state): the baseline, the counterfactual with the labor-supply response shut off, and the counterfactual with selection-neutral layoffs. The gap from the baseline to each counterfactual is that channel's contribution.

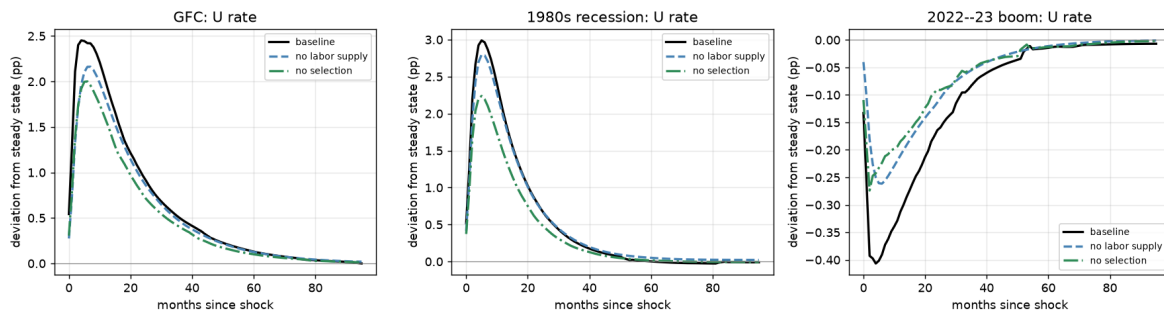


Figure 13: **Labor supply and selection in unemployment, by episode.** Unemployment rate response of each episode for the counterfactuals in Figure 12.

The episodes also sharpen the measurement lesson of Section 10. In the late 1970s, the marginal workforce was bigger, yet unemployment and WTW track each other more closely over the 1980s cycle because the modest UI policy introduced little wedge between searching and wanting to work. In the 2022-23 boom, they diverge for the opposite reason: the unemployment rate falls 0.4 points while the WTW share falls a full point and employment rises 1.1: the tight market absorbed jobless workers, most from outside measured unemployment, exactly the margin the unemployment rate misses. Across episodes, the unemployment rate is a good proxy for slack precisely when UI policy is quiet and the marginal workforce is

small; when either condition fails, the flows we construct— in particular, the destination of laid-off workers— are the observable correction.

12. Conclusion

In this paper, we have made the case for a revised workhorse model of labor markets over the business cycle. Confronted with new monthly CPS series on quits and layoffs by destination, the standard model fails because it assumes all layoffs are random and all quits are selective. By introducing layoffs that select on employment surplus and quits that strike workers far from the participation margin, we improve the standard framework to successfully match the empirical composition of separations over the business cycle. These additions also bring its predictions in line with micro-level evidence on cyclical selection (Mueller (2017)) and pass other validation tests from the PSID and CPS.

Getting this composition right changes our understanding of recessionary dynamics by introducing or strengthening two channels. First, labor supply stabilizes the cycle: marginal workers hoard jobs, and displaced workers stay attached. These two phenomena buffer the employment decline while amplifying measured unemployment. Second, output per worker falls not only through technology shocks but also through composition, which cuts the estimated welfare cost of the GFC recession by more than half. Both channels induce a greater share of the non-employed to actively search in unemployment instead of passively searching in non-participation during recessions, making the official unemployment rate a noisier gauge of slack precisely when it is watched most closely. We make the case that to follow idle capacity in near-real time, researchers should track the easily available data on the jobless who *want to work* WTW in the CPS.

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A. Data Robustness and Validation

A.1. Robustness

We provide robustness checks in this section around the main data facts in the paper: level and cyclicalities of quits as well as level and cyclicalities of share of layoffs into non-participation. For a detailed insight into robustness of the entire data, please see our paper, [Ellieroth and Michaud \(2026\)](#), in which we construct and analyze the full data series.

The question used to separate the flows from employment to non-participation by reason is only asked to individuals in the outgoing rotation groups. Due to that, a few possible issues might arise: (ii) there might be differences between the two outgoing rotation groups (month-in-sample 4 and 8), (iii) individuals in the outgoing rotation groups might be different than the other month-in-sample groups, and lastly, (iv) there is some non-response among the people asked in the outgoing rotation groups which could cause results to be biased. In addition, (v) we show that deNUNifying the data does not change our findings either. DeNUNification means dropping what could be spurious transitions, namely individuals who move between N and U, i.e individuals with either of the following transition patterns: N-U-N or U-N-U.

Table 15 shows that the main findings motivating this paper are robust to the aforementioned potential biases due to selection, non-response, and misclassification. The first row shows the numbers shown and used in this paper. Rows 2 through 4 show the numbers when we reweigh the data to make it representative. The second row shows the numbers when we test for differences between the two outgoing rotation groups. Since we do not find any statistically significant differences between the two ORGs, the results using weights are the same. Row 3 shows the numbers when reweigh the data so that the individuals in the ORGs are representative of everyone in a given month. Again, since the individuals in the ORGs are statistically the same as in the other month-in-sample (MIS) groups, the results are identical when applying weights. The only row where we observe slight differences is correcting for non-response, the cyclicalities of quits becomes slightly more countercyclical. Lastly, deNUNification of the data yields the exact same results as in the baseline.

	Level	Quits Cyclicalities	Level	Share of Layoffs into N Cyclicalities
(1) Baseline	0.85	-0.46	0.35	-0.62
(2) ORG 4 vs. 8	0.85	-0.46	0.35	-0.62
(3) ORG vs. other MIS	0.85	-0.46	0.35	-0.62
(4) Non-Response among ORG	0.85	-0.48	0.35	-0.62
(5) DeNUNification	0.85	-0.46	0.35	-0.62

Table 15: Robustness Results

B. Panel Study of Income Dynamics (PSID)

The PSID is a long-running panel survey that has grown to over 9,000 families. While the smaller sample size limits the accuracy of business cycle analysis in the PSID relative to CPS or JOLTS, the comprehensiveness of the survey surpasses the other two sources. Our sample is PSID heads and spouses/partners in the 2003-2021 surveys and follows the construction in [Ellieroth and Michaud \(2026\)](#), re-explained here. Surveys are retrospective, so responses cover 2001-2019. Similar to the CPS, we construct a panel with both the reason a job ended together with the reported year and month of the separation. These items are collected separately for the household head and for the spouse/partner for up to four jobs held in the year.

The classification of reason for separation uses the question on why a job ended (e.g. variable ER25431 for the spouse in the 2005 wave). The question is interviewer-coded from an open response prompted as follows:

DE51. Why did she stop working for (NAME OF EMPLOYER)? Did the company go out of business, was she laid off, did she quit, or what?

Responses are coded into the categories reproduced in Table 16.

Placing each separation on a monthly timeline requires the month in which the job ended (variables hhMoEnd/hwMoEnd), together with the end year. A monthly separation date is formed as the year-month combination for every observation classified as a quit, layoff, or temporary-job ending, and duplicate person-months are removed so that each individual contributes at most one separation per calendar month.

We merge the dated separations onto a monthly employment panel (providing monthly employment, unemployment, and out-of-labor-force status and the number of jobs held). Using the labor-market state in the separation month and the following month, each quit and each layoff is assigned to one of three mutually exclusive destinations: out of the labor

Code	Text	Classification
1	Company folded / changed hands / moved; employer died / out of business	Layoff
2	Strike; lockout	Dropped
3	Laid off; fired	Layoff
4	Quit; resigned; retired; pregnant; needed more money; wanted a change	Quit
7	Other; transfer; any mention of armed services	Dropped
8	Job was completed; seasonal work; was a temporary job	Layoff
9	DK; NA; refused	Dropped
0	Inap. (no such person / did not work / still with employer, etc.)	Dropped

Table 16: Response codes for the “why last job ended” item

force (N), unemployment (U), or employment (E). We drop direct E-to-E transitions since we cannot see them in the CPS and our paper studies only E-to-Non-Employment transitions. This yields four categories: QN , QU for quits, and LN , LU for layoffs. Raw observation counts are shown in 17 but all other tabulations use PSID longitudinal sampling weights.

<i>Panel A: Quits</i>	
Out of labor force (QN)	2,968
Unemployed (QU)	423
<i>Panel B: Layoffs</i>	
Out of labor force (LN)	1,602
Unemployed (LU)	2,921

Table 17: PSID: Cell counts underlying prime-age tabulations (ages 25–55)

Table 18 shows that the split between quits and layoffs to non-employment that end up in non-participation is similar in the PSID as it is in our CPS sample. Over the same period, 15% of prime age quits to non-employment are classified as unemployment in the CPS compared to 12% in the PSID sample; and 65% of prime age layoffs to non-employment are to unemployment compared to 62.0% in the CPS sample.¹⁷ This provides confidence that our classification of quits and layoffs is consistent with how workers describe the reason for a job ending in other popular surveys.

¹⁷Quits to unemployment with an unemployment duration of over one year are dropped.

	Share conditional on non-employment			
	CPS		PSID	
	All	Rec.	All	Rec.
<i>Panel A. Quits</i>				
to N (QN)	0.85	0.84	0.88 (0.008)	0.88 (0.020)
to U (QU)	0.15	0.16	0.12 (0.008)	0.12 (0.020)
Total	1.00	1.00	1.00	1.00
<i>Panel B. Layoffs</i>				
to N (LN)	0.35	0.31	0.38 (0.013)	0.28 (0.021)
to U (LU)	0.65	0.69	0.62 (0.013)	0.72 (0.021)
Total	1.00	1.00	1.00	1.00

Table 18: **Share of separations flowing to each destination (2001-2019).** Destinations are (N)on-participation or (U)nemployment. Types of separation are (Q)uit or (L)ayoff. The table conditions on separations into non-employment. “Rec.” restricts to recession months. Columns sum to one within each panel. Standard errors for the PSID are clustered on the individual and reported in parentheses.

B.1. PSID Kernel Densities’ Construction.

This section details the construction of the kernel density estimates for non-employment spell durations (Figure 4 and pre-separation residual earnings (Figure 3.

Non-Employment Spell Durations. To construct the distributions of non-employment durations, we restrict the sample to completed non-employment spells lasting fewer than 48 months. We compute the probability density function for duration d conditional on the initiating separation type $s \in \{QN, LN, LU\}$ using a standard kernel density estimator with bandwidth of $h = 2$ months across all transition groups to ensure comparability. The distributions are plotted separately for Quits to Non-participation (QN), Layoffs to Non-participation (LN), and Layoffs to Unemployment (LU).

Pre-Transition Residual Earnings To analyze unobserved worker heterogeneity prior to separation, we estimate the distribution of residual earnings in the period leading up to a labor market transition. The sample is restricted to prime-age workers (ages 25 to 54). To mitigate the influence of extreme outliers, we further condition on pre-transition monthly earnings w_{it} such that $1000 < w_{it} < 10000$.

We first remove deterministic life-cycle and educational earnings profiles by regressing raw earnings on a college education indicator fully interacted with a cubic polynomial in age. The specification is given by:

$$w_{it} = \sum_{c \in \{0,1\}} \mathbb{I}(Col_i = c) (\beta_{0,c} + \beta_{1,c} Age_{it} + \beta_{2,c} Age_{it}^2 + \beta_{3,c} Age_{it}^3) + \varepsilon_{it}$$

where w_{it} represents the earnings of individual i at month t , $\mathbb{I}(Col_i = c)$ is an indicator variable for college degree attainment, and ε_{it} captures the unobserved, residual component of earnings.

We then plot the kernel density estimates of the predicted residuals of the regression for observations exactly three months prior to an observed separation ($t + 3$). As with the duration distributions, the densities of the earnings residuals are disaggregated by the subsequent transition type (QN, LN, and LU).

C. Computational Appendix

This appendix details the algorithm summarized in the Computation and Estimation sections of the main text. The period is one month; all quantities are monthly.

C.1. State space and discretization

The individual state is (a, z, p, e) : assets a , idiosyncratic productivity z , the participation-constraint indicator p , and (for the unemployed) UI eligibility e . Assets lie on a grid of $N_a = 48$ points spanning $[0, 30]$ in units of the median monthly wage, with quadratic spacing ($a_i = 30(i/47)^2$) so points concentrate near the borrowing constraint where the consumption policy is most curved. Log productivity follows $\log z' = \rho_z \log z + \varepsilon$, $\varepsilon \sim N(0, \sigma_\varepsilon^2)$, discretized on $N_z = 15$ points by the method of [Tauchen \(1986\)](#): the grid spans ± 3 unconditional standard deviations and σ_ε^2 is set so the stationary variance of $\log z$ equals 0.20 at the estimated ρ_z . We carry five labor-market “statuses,” namely employed (E), unemployed without and with UI eligibility (U_0, U_1), and free and participation-constrained non-participants (N_F, N_C), so that eligibility and the participation constraint are tracked as discrete states rather than approximated. The continuation-cost (selective-layoff) shock arrives i.i.d. with probability $\lambda_x(Z)$; the participation process moves workers between the free and constrained non-participation states with persistence (p_0, p_1) ; eligibility is lost at rate $\mu(Z)$.

C.2. Stationary equilibrium

For a candidate labor tax τ we solve the five value functions by value-function iteration to a sup-norm tolerance of 10^{-7} . Each backward step forms, for every status, the expected continuation value $\mathbb{E}_{z'|z}[C_s(a', z')]$ from next period’s value functions (where C_s embeds the discrete

separation, job-offer, eligibility, and participation lotteries described in Section 7) and then maximizes $u(c) - \theta \mathbf{1}\{\text{search}\} + \beta(1 - R_d) \mathbb{E}[C_s]$ over a' subject to the budget constraint with the consumption floor $\underline{c}$ imposed. The savings policy is monotone in a within each (s, z) , which we exploit so the maximization is $O(N_a)$ rather than $O(N_a^2)$. The invariant distribution over (s, a, z) is computed by non-stochastic iteration of the implied transition operator to a tolerance of 10^{-11} ; because the operator is applied exactly, the resulting moments carry *no* simulation noise, which is what lets the SMM objective be optimized by derivative-based and simplex methods without Monte-Carlo error. Alongside the distribution we iterate a tagged sub-distribution D^{sel} that records, in each non-employed cell, the mass whose current non-employment spell began with a *selective* separation (a selective quit or a selective layoff); the ratio D^{sel}/D delivers the stock-selectivity diagnostics in Table 9. Newborns (mass R_d) replace the deceased, enter non-employed holding average assets, draw z from its stationary distribution, and are counted as originating from non-participation. Finally, the labor tax is the fixed point of the period-by-period government budget, $\tau = (\text{UI outlays} + \text{floor outlays})/\text{wage bill}$, found by damped iteration. Prices come from the capital-effective-labor ratio $\kappa = K/L$: $r = r^* A(Z)(\kappa/\kappa^*)^{\alpha_y - 1}$ and $w = A(Z)(\kappa/\kappa^*)^{\alpha_y}$, with K aggregate household savings and L aggregate efficiency units of employed labor, normalized so $w^* = 1$ and r^* delivers a one-percent annual return.

C.3. Business cycles as a vector impulse

A recession is a single unanticipated MIT vector impulse, following Boppart et al. (2018). Rather than holding the cyclical parameters at their bad-state values for a fixed spell, we let the business cycle be one innovation that propagates with persistence: at the impact month every cyclical parameter $\Theta \in (A, \delta, \lambda_u, \lambda_n, \lambda_x, \mu)$ jumps to its bad-state value, and thereafter its deviation from the normal-times value decays geometrically at the common rate ρ_x ,

$$\Theta_t = \Theta^G + (\Theta^B - \Theta^G) \rho_x^{t-1}, \quad t = 1, 2, \dots,$$

with $\Theta_0 = \Theta^G$. The economy is computed forward for a total of $T = 300$ months to approximate return to the steady state. We set $\rho_x = 0.95$ at the monthly frequency. The episodes of Section 11 use lower ρ_x values that match their own, faster persistence (0.92 for the 1981–82 recovery, 0.94 for the 2022–23 boom).¹⁸ Given guesses for the paths of κ_t (hence prices) and

¹⁸We experimented with a two-timescale impulse, a slowly-decaying aggregate (TFP) component together with fast-decaying separation/finding shocks, to recover the data’s persistence without sacrificing the cross-sectional behavior. It does not help, for an instructive reason. Technology moves the unemployment rate almost not at all (Section 6), so a persistent TFP component does not lengthen the unemployment gap; the gap’s persistence is governed by the persistence of the job-finding collapse. But a persistent finding collapse, while it preserves hoarding, makes non-participation a dead end and collapses the share of layoffs flowing there, whereas a fast-recovering finding rate restores that share but removes the reason to hoard.

the budget-clearing tax τ_t , we solve the household problem by backward induction from the steady-state value functions as the terminal condition, iterate the distribution forward from the steady-state distribution, and update (κ_t, τ_t) by damped fixed point (relaxation weight 0.5) until the path error falls below 2×10^{-4} ; convergence is fast because prices and taxes move little over the cycle. A moment’s recession “trough” is its largest absolute deviation from steady state along the impulse; under the geometric impulse the slow-moving stocks reach their troughs well after the one-month forcing, so we search a 180-month window. Recession averages are taken over the first 48 months. The volatility ratio $\text{std}(L)/\text{std}(Y)$ uses a 120-month window of log deviations, and the GDP decline is the peak-to-trough fall in $Y_t = A_t K_t^{\alpha_y} L_t^{1-\alpha_y}$ relative to its steady-state level.

C.4. How decomposition counterfactuals are constructed.

Three counterfactuals isolate the channels. (i) The *labor-supply* role freezes savings rules, discrete decision rules, and prices at their steady-state values along the recession path, while feeding in the *baseline* incidence of selective layoffs (the realized stay-vs-separate decisions), so the role captures only the quit, search, and participation responses and not changes in who is laid off. (ii) The *selection* role re-solves the recession’s (δ^B, λ_x^B) so that the share of layoffs to non-participation stays at its steady-state value while the layoff-rate path matches the baseline (a two-equation root-find on window-average sN and window-average layoff rate), and compares the resulting transition to the baseline. (iii) The *one-shock-at-a-time* decomposition switches each cyclical parameter to its bad-state value alone, holding the others at normal, with agents understanding each path.

C.5. Estimation

We estimate by Simulated Method of Moments, minimizing a weighted sum of squared proportional deviations of the model moments from their targets, stacked over the normal and recession states, plus the GDP-decline term. The weights are those given in the Estimation section: the employment-population ratio and unemployment rate carry the heaviest weights, the new separation series and attachment flows intermediate weights, and the basic gross flows the least. Because steady-state moments do not depend on the recession parameters, we estimate in two stages (stage A on the ten steady-state parameters and the normal-times moments, stage B on the five recession parameters and the recession deviations) and then polish jointly over all fifteen by multi-start Nelder–Mead, finishing by bisectioning A^B so the trend-adjusted GDP decline is matched exactly with the other recession parameters

Unemployment persistence, hoarding, and the destination of layoffs are all governed by the same finding-rate clock and pull against one another; no decay structure delivers all three, and $\rho_x = 0.95$ is the scalar compromise.

re-optimized around it.

C.6. Welfare

For each agent at the steady-state distribution we compute the permanent consumption-equivalent λ that equates the value of remaining at the steady state to the value of entering the recession at date zero, $\lambda = \exp\{(V^{ss} - V^{rec})(1 - \beta(1 - R_d))\} - 1$, where the annuitization factor $1 - \beta(1 - R_d)$ is the effective per-period discount including mortality. We report the mean and the breakdown by labor-market state at the recession's onset. The UI-extension experiment replaces $\mu^B = 0$ with the normal-times expiry rate μ^G and re-solves the transition; the selection-neutral welfare numbers use the counterfactual of the previous subsection.

D. Identification, Standard Errors, and Sensitivity

This appendix formalizes the verbal argument of Section 5.1. Let $m(\theta)$ stack the normal-times and recession values of the eleven targeted moments and let θ collect the parameters governing labor supply (the search disutility θ and home production h) and the good- and bad-time frictions (the offer rates λ_u, λ_n , the random layoff rate δ , and the continuation-cost arrival λ_x). We compute the moment Jacobian $J = \partial m / \partial \theta$ by central finite differences at the estimated point, with steps scaled to each parameter.

Standard errors. Table 19 reports SMM standard errors from the sandwich formula $\text{Var}(\hat{\theta}) = (J'WJ)^{-1}J'W\Omega WJ(J'WJ)^{-1}$, where W is the estimation weighting matrix and Ω is a diagonal estimate of the targets' sampling variances, formed from the month-to-month variance of each CPS series over the steady-state and recession windows divided by the number of months. The parameters are precisely estimated. Most informative for this paper, the continuation-cost arrival λ_x and the participation parameters, the two ingredients that repair the model, carry large t -statistics, confirming that the separation-composition and attachment moments identify them rather than leaving them free.

Sensitivity. Table 20 reports the local sensitivity matrix $S = (J'WJ)^{-1}J'W$ of Andrews et al. (2017), expressed as elasticities: entry (k, j) is the percent change in parameter k induced by a one-percent change in target j . The pattern confirms the verbal mapping of Section 5.1. The share of layoffs to non-participation loads most heavily on the continuation-cost arrival λ_x (and, through the estimated threshold, on the cost x): it is the moment that identifies selection. The quit rate and the non-participation re-entry rate NU load on the participation parameters (p_0, p_1) , which split quits into their marginal and detached components. Home production h and the search disutility θ are identified jointly by the employment-population

Table 19: Identification: estimates, SMM standard errors, and t -statistics for the parameters governing labor supply (search disutility, home production) and the good- and bad-time frictions (offer rates, random layoff, continuation-cost shock). Standard errors use the sandwich formula with the estimation weighting matrix and diagonal target sampling variances from the monthly CPS series.

Parameter	Estimate	Std. error	t -stat
θ (disutility of search)	0.0953	0.0226	4.2
h (home production)	0.5119	0.0034	151.4
λ_u^G (offer rate U, good)	0.2733	0.0025	107.5
λ_n^G (offer rate N, good)	0.1119	0.0018	61.8
δ^G (random layoff, good)	0.0086	0.0002	34.7
λ_x^G (cont.-cost shock, good)	0.0857	0.0036	24.1
λ_u^B (offer rate U, bad)	0.2239	0.0073	30.5
λ_n^B (offer rate N, bad)	0.0610	0.0038	16.0
δ^B (random layoff, bad)	0.0152	0.0013	11.4
λ_x^B (cont.-cost shock, bad)	0.0921	0.0061	15.2

ratio, the quit level, and the destination of layoffs: the moments that locate the mass of workers near the participation margin.

Table 20: Local sensitivity of estimated parameters to the normal-times target moments: elasticities from the sensitivity matrix $S = (J'WJ)^{-1}J'W$. An entry is the percent change in the row parameter induced by a one-percent change in the column target.

Parameter	epop	urate	EU	EN	UE	NE	quit	layoff	share_layoff_N	UN	NU
θ (disutility of search)	13.24	-0.35	-0.00	-0.00	-0.38	-0.89	-0.00	-0.02	-0.73	-1.51	0.00
h (home production)	-1.24	-0.08	-0.00	0.00	0.09	-0.00	0.00	-0.00	-0.01	0.00	-0.00
λ_u^G (offer rate U, good)	1.27	0.07	0.00	0.00	0.76	0.07	0.00	0.00	0.07	-0.07	-0.00
λ_n^G (offer rate N, good)	0.14	-0.10	0.00	0.00	0.68	0.29	0.00	0.01	-0.14	0.26	-0.01
δ^G (random layoff, good)	-8.72	0.41	0.00	0.00	0.64	0.18	0.00	0.01	-0.76	0.02	-0.00
λ_x^G (cont.-cost shock, good)	-4.28	0.77	0.00	-0.00	0.03	-0.04	-0.00	0.00	1.49	-0.19	0.01
λ_u^B (offer rate U, bad)	-1.90	-0.17	-0.00	-0.00	0.14	-0.05	0.00	-0.00	-0.08	-0.03	0.00
λ_n^B (offer rate N, bad)	-5.96	-0.47	-0.00	0.00	-0.22	0.02	0.00	-0.00	-0.44	0.31	0.00
δ^B (random layoff, bad)	8.30	-0.23	-0.00	-0.00	-0.11	-0.18	-0.00	-0.01	-0.32	-0.19	0.00
λ_x^B (cont.-cost shock, bad)	20.48	1.14	0.00	-0.00	-0.87	0.03	-0.00	0.01	0.24	0.06	0.01

Recession accounting with standard deviations instead of steady-state to trough changes. Table 21 reports the sources of recessions decomposition in volatility terms, as the standard deviation each shock generates on its own.

Table 22 reports the labor-supply and selection roles in volatility terms.

Moment	All shocks	TFP	Job finding	Job loss	UI extension
<i>Full model</i>					
E/pop	1.03	0.00	0.45	0.54	0.09
U rate	14.29	0.05	5.30	8.10	2.11
UE rate	3.91	0.00	3.91	0.00	0.00
UN rate	7.08	0.35	7.56	2.73	8.03
NU rate	10.26	0.18	11.87	0.71	0.89
Quit rate	8.58	0.00	12.10	0.39	0.06
Layoff rate	8.41	0.00	1.01	9.77	0.57
Share of layoffs to N	8.04	0.00	1.47	4.84	1.17
<i>Standard model</i>					
E/pop	0.91	0.25	0.08	0.50	0.07
U rate	11.02	1.17	1.95	6.06	3.60
UE rate	0.60	0.00	0.60	0.00	0.00
UN rate	16.59	2.16	2.56	0.36	16.54
NU rate	8.61	6.75	15.03	3.85	15.76
Quit rate	1.80	2.48	4.25	0.73	0.54
Layoff rate	6.69	0.00	0.00	6.69	0.00
Share of layoffs to N	4.46	3.31	3.42	0.21	2.33

Table 21: What drives recessions, in volatility terms. Each entry is the standard deviation of the HP-filtered log of the moment (percent) generated along the impulse when only that shock is active.

Moment	Full model			Standard model	
	Baseline	Labor supply	Selection	Baseline	Labor supply
E/pop	1.03	-0.39	-0.31	1.16	+0.44
U rate	14.29	+1.86	+2.66	12.22	+0.57
UN rate	7.08	-1.62	+2.88	15.98	+0.78
NU rate	10.26	+9.18	+4.50	11.72	+6.51
Quit rate	8.58	+7.17	+3.15	7.98	+7.64
Layoff rate	8.41	+0.74	-0.24	8.80	+0.00
Share of layoffs to N	8.04	-0.44	+7.28	2.97	+2.75
Want-to-work share	4.96	-0.55	-0.58	4.34	+1.20

Table 22: **Roles of labor supply and selection, in volatility terms.** “Baseline” is the standard deviation of the HP-filtered log of the moment (percent) along the impulse; “Labor supply” and “Selection” are how much of that volatility each channel adds, measured as the baseline volatility minus the volatility with that channel switched off.